MAT1001

Calculus I

Homework Solutions

L07 • Fall 2026

Sections 5.1–16.4 • 337 assigned exercises • 27 sections

使用说明

按课程作业清单编排，保留教材原题号和指定小题。

每道题先标注解法相近的教材例题，再给出中文思路与英文解答。

教材以图例、定义或定理讲解该方法时，题前标注相应出处。

7.8 第 21 题收录 (a)。16.1 第 18 题可在学完 7.4 和 16.2 后阅读。

Thomas' Calculus in SI Units, 14th Edition

Scope and order follow MAT1001 L07 Course Details 2026.

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5.1 Area and Estimating with Finite Sums

Assigned: 1, 3, 5, 7, and 19 • Thomas' Calculus, 14th SI edition, pp. 274–276

METHOD | RECTANGULAR SUMS

Divide the interval into equal widths $\Delta x = (b-a)/n$.
For lower and upper sums, use the minimum and maximum height on each subinterval.
For the midpoint rule, use the height at the center of each subinterval.

Exercises 1 and 3: Lower and Upper Sums

Find the lower sums with two and four rectangles, then the upper sums with two and four.

参考教材例题 1（第 270 页，端点和）

Exercise 1

$f(x) = x^2$ on $[0, 1]$

解题思路

函数在区间上递增，所以每个小区间的左端点给出最小高度，右端点给出最大高度。先用两个等宽矩形，再用四个；每次都把矩形的面积相加。

SOLUTION

Since f is increasing on $[0, 1]$, use left endpoints for the lower sums and right endpoints for the upper sums. The widths are $1/2$ and $1/4$, respectively.

(a) $L_2 = \frac{1}{2}\left(0 + \frac{1}{4}\right) = \frac{1}{8}.$

(b) $L_4 = \frac{1}{4} \cdot \frac{0+1+4+9}{16} = \frac{7}{32}.$

(c) $U_2 = \frac{1}{2}\left(\frac{1}{4} + 1\right) = \frac{5}{8}.$

(d) $U_4 = \frac{1}{4} \cdot \frac{1+4+9+16}{16} = \frac{15}{32}.$

Hence the four-rectangle bounds on the area A are

$$\frac{7}{32} \le A \le \frac{15}{32}.$$

参考教材例题 1（第 270 页，端点和）

Exercise 3

$f(x) = 1/x$ on $[1, 5]$

解题思路

函数在正数区间递减，端点的角色与第 1 题相反：右端点给下和，左端点给上和。两次等分的矩形宽度分别是 2 和 1。

SOLUTION

Since f is decreasing on $[1, 5]$, use right endpoints for the lower sums and left endpoints for the upper sums. The widths are 2 and 1, respectively.

(a) $L_2 = 2\left(\frac{1}{3} + \frac{1}{5}\right) = \frac{16}{15}.$

(b) $L_4 = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{77}{60}.$

(c) $U_2 = 2\left(1 + \frac{1}{3}\right) = \frac{8}{3}.$

(d) $U_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12}.$

Hence the four-rectangle bounds on the area A are

$$\frac{77}{60} \le A \le \frac{25}{12}.$$

Method reference: Section 5.1, Figs. 5.2–5.3 and Example 1 (textbook pp. 267–270).

Exercises 5 and 7: The Midpoint Rule

Estimate each area using first two, then four equal-width rectangles with midpoint heights.

参考教材图 5.3(b)（第 268 页，中点法）

Exercise 5

$f(x) = x^2$ on $[0, 1]$

解题思路

先找每个小区间的中点，再用中点横坐标的平方作矩形高度。同一次等分的宽度相同，可以提到和式外。

SOLUTION

With two rectangles of width $1/2$, the midpoints are $1/4$ and $3/4$. Thus

$$M_2 = \frac{1}{2} \cdot \frac{1+9}{16} = \frac{5}{16} = 0.3125.$$

With four rectangles of width $1/4$, the midpoints are $1/8$, $3/8$, $5/8$, and $7/8$. Thus

$$M_4 = \frac{1}{4} \cdot \frac{1+9+25+49}{64} = \frac{21}{64} = 0.328125.$$

参考教材图 5.3(b)（第 268 页，中点法）

Exercise 7

$f(x) = 1/x$ on $[1, 5]$

解题思路

和第 5 题一样取中点。这次矩形高度是中点的倒数。算面积时还要乘以矩形宽度；用两个矩形时宽度为 2。

SOLUTION

For width 2, the midpoints are 2 and 4, so

$$M_2 = 2\left(\frac{1}{2} + \frac{1}{4}\right) = \frac{3}{2} = 1.5.$$

For width 1, the midpoints are $3/2$, $5/2$, $7/2$, and $9/2$, so

$$M_4 = \frac{2}{3} + \frac{2}{5} + \frac{2}{7} + \frac{2}{9} = \frac{496}{315} \approx 1.574603.$$

Estimation from Rate Data

参考教材例题 1（第 270 页，由速率估计累计量）

Exercise 19

Water pollution

解题思路

把每小时的漏油速率乘以 1 小时，再把各小时的估计漏油量相加。按题意把速率视为递增，左端点给下界，右端点给上界。前 8 小时漏得越多，剩余油量越少，漏完也就越早。

SOLUTION

Let $q(t)$ be the leak rate in L/h. As in Example 1, multiply each sampled rate by the subinterval width, here 1 h. Assuming q increases between readings, the left-endpoint sum L_n is a lower bound, and the right-endpoint sum U_n is an upper bound on the amount lost in n hours.

(a) For the first five hours,

$$L_5 = (50 + 70 + 97 + 136 + 190)\,\text{L} = 543\,\text{L},$$

$$U_5 = (70 + 97 + 136 + 190 + 265)\,\text{L} = 758\,\text{L}.$$

(b) Add the contributions for the next three hours:

$$L_8 = L_5 + (265 + 369 + 516)\,\text{L} = 1693\,\text{L},$$

$$U_8 = U_5 + (369 + 516 + 720)\,\text{L} = 2363\,\text{L}.$$

(c) If Q_8 liters leaked in the first eight hours, then at the constant rate of 720 L/h the time remaining is $(25{,}000 - Q_8)/720$ hours. This decreases as Q_8 increases. The earliest full spill (worst case) is estimated at

$$\frac{25{,}000 - 2363}{720}\,\text{h} \approx 31.4\,\text{h}.$$

The latest full spill (best case) is estimated at

$$\frac{25{,}000 - 1693}{720}\,\text{h} \approx 32.4\,\text{h}.$$

5.2 Sigma Notation and Limits of Finite Sums

10 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 282–283

参考教材例题 1，第 277 页

Exercise 8

解题思路

比较各个和式的首项、项数和公比。

SOLUTION

In (a) set $j = k - 1$ and in (b) write $(-1)^k 2^k = (-2)^k$. Both give $\sum_{j=0}^{5}(-2)^j$. In (c) the first term is -1 and the six terms alternate with ratio -2, so (c) has the opposite sign. Thus (a) and (b) give the stated sum.

参考教材例题 2，第 277 页

Exercise 12

解题思路

把每一项写成连续正整数的平方。

SOLUTION

The four terms are k^2 for $k = 1, 2, 3, 4$. Hence the sum is $\sum_{k=1}^{4} k^2$.

参考教材例题 4，第 279 页

Exercise 30

解题思路

先用幂和公式从 1 加起，再减去范围外的项。

SOLUTION

Write $S_1(n) = n(n+1)/2$ and $S_2(n) = n(n+1)(2n+1)/6$.

(a) $S_1(36) - S_1(8) = 630$.

(b) $S_2(17) - S_2(2) = 1780$.

(c) Since $k(k-1) = k^2 - k$, the sum is $S_2(71) - S_2(17) - S_1(71) + S_1(17) = 117648$.

参考教材例题 3–4，第 278–279 页

Exercise 32

解题思路

与指标无关的项直接乘项数，含指标的项用一次幂和。

SOLUTION

For (a) and (b) each term is constant in k, giving $n(1/n + 2n) = 1 + 2n^2$ and $n(c/n) = c$, respectively. For (c), $\sum_{k=1}^{n} k/n^2 = (n+1)/(2n)$.

参考教材例题 5，第 279 页；图 5.9，第 281 页

Exercise 37

解题思路

四个矩形等宽。分别在左端、右端和中点取样，负的函数值对应轴下的矩形。

SOLUTION

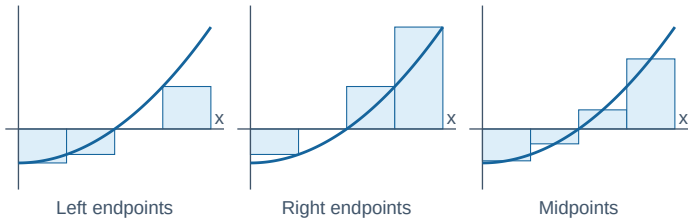
The partition points are $0, \frac{1}{2}, 1, \frac{3}{2}, 2$. The heights, from left to right, are:

(a) $-1, -\frac{3}{4}, 0, \frac{5}{4}$.

(b) $-\frac{3}{4}, 0, \frac{5}{4}, 3$.

(c) $-\frac{15}{16}, -\frac{7}{16}, \frac{9}{16}, \frac{33}{16}$.

The three sketches use these heights on bases of width $1/2$.



参考教材例题 5，第 279 页；图 5.9，第 281 页

Exercise 39

解题思路

把区间四等分，利用单位圆求各取样点的正弦。

SOLUTION

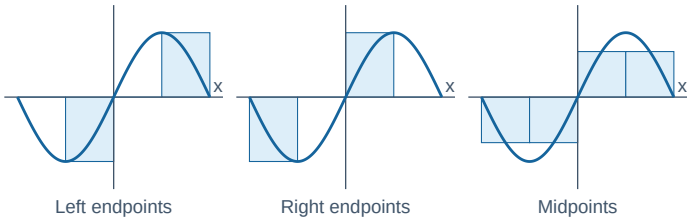
The partition points are $-\pi, -\pi/2, 0, \pi/2, \pi$. Put $r = \sqrt{2}/2$. The heights are:

(a) $0, -1, 0, 1$.

(b) $-1, 0, 1, 0$.

(c) $-r, -r, r, r$.

Each rectangle has base width $\pi/2$.



参考教材例题 6，第 282 页

Exercise 41

解题思路

作相邻分点的差，取最大间距。

SOLUTION

The subinterval lengths are $1.2, 0.3, 0.8, 0.3, 0.4$. Their maximum is $\|P\| = 1.2$.

参考教材例题 6，第 282 页

Exercise 42

解题思路

用右端点减左端点，尤其留意负号。

SOLUTION

The successive widths are $0.4, 1.1, 0.5, 0.8, 0.2$. Hence $\|P\| = 1.1$.

参考教材例题 5，第 279 页

Exercise 43

解题思路

把右端点代入二次函数，用平方和公式化简后取极限。

SOLUTION

Here $\Delta x = 1/n$ and $x_k = k/n$. Thus

$$S_n = 1 - \frac{(n+1)(2n+1)}{6n^2}.$$

Indeed, the subtracted term is $\sum_{k=1}^{n} k^2/n^3$. The limit, and hence the area, is $1 - \frac{1}{3} = \frac{2}{3}$.

参考教材例题 5，第 279 页

Exercise 44

解题思路

用等宽右端点和，把计算化为连续整数求和。

SOLUTION

Here $\Delta x = 3/n$ and $x_k = 3k/n$, so

$$S_n = \frac{18}{n^2}\sum_{k=1}^{n} k = \frac{9(n+1)}{n}.$$

As $n \to \infty$, this tends to the area 9.

5.3 The Definite Integral

15 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 292–294

参考教材定积分的定义，第 284 页

Exercise 1

解题思路

从求和项辨认函数，从分割区间读出积分上下限。

SOLUTION

The summand is $f(c_k)\Delta x_k$ with $f(x) = x^2$. By continuity the limit is $\int_0^2 x^2\,dx$.

参考教材定积分的定义，第 284 页

Exercise 3

解题思路

把取样点换成积分变量，保留原区间。

SOLUTION

The integrand is $f(x) = x^2 - 3x$, continuous on $[-7, 5]$. The limit is $\int_{-7}^{5}(x^2 - 3x)\,dx$.

参考教材定积分的定义，第 284 页

Exercise 6

解题思路

根式就是被积函数，区间端点给出上下限。

SOLUTION

The continuous function sampled at c_k is $f(x) = \sqrt{4 - x^2}$ on $[0, 1]$. The limit is $\int_0^1 \sqrt{4 - x^2}\,dx$.

参考教材定积分的定义，第 284 页

Exercise 8

解题思路

辨认正切函数，并确认它在分割区间上连续。

SOLUTION

Since $\tan x$ is continuous on $[0, \pi/4]$, the limit is $\int_0^{\pi/4} \tan x\,dx$.

参考教材例题 2，第 289 页

Exercise 9

解题思路

分别用零长区间、换向、线性和区间可加性。

SOLUTION

The six values follow from the integral rules:

(a) 0, since the limits agree.

(b) -8, by reversing the limits.

(c) $3(-4) = -12$.

(d) $6 - (-4) = 10$.

(e) $6 - 8 = -2$.

(f) $4(6) - 8 = 16$.

参考教材例题 5，第 292 页

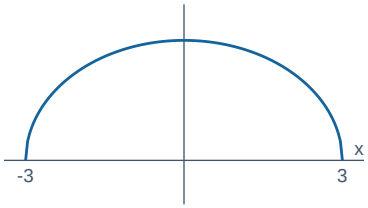
Exercise 17

解题思路

图形是以原点为圆心、半径为 3 的上半圆。

SOLUTION

The graph $y = \sqrt{9 - x^2}$ is the upper half of $x^2 + y^2 = 9$. Thus the integral equals its area, $\frac{1}{2}\pi(3)^2 = 9\pi/2$.



参考教材例题 4，第 289 页

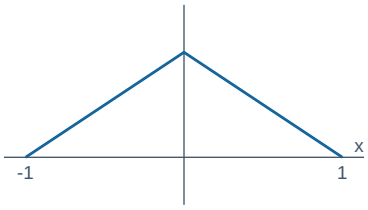
Exercise 20

解题思路

绝对值图形在零点折转，积分是三角形面积。

SOLUTION

The graph joins $(-1, 0), (0, 1), (1, 0)$ by line segments. The region is a triangle of base 2 and height 1, so the integral is $\frac{1}{2}(2)(1) = 1$.



参考教材例题 5，第 292 页

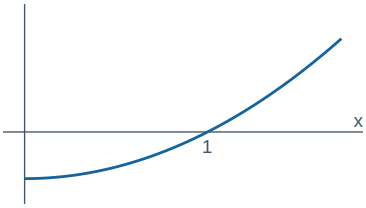
Exercise 55

解题思路

先辨认抛物线和零点，再用积分除以区间长度。

SOLUTION

The graph is the increasing branch of $y = x^2 - 1$ from $(0, -1)$ through $(1, 0)$ to $(\sqrt{3}, 2)$. By Eq. (4), $\int_0^{\sqrt{3}} x^2\,dx = (\sqrt{3})^3/3 = \sqrt{3}$. Subtracting the integral of 1 gives zero, so the average is 0.



参考教材表 5.6，第 288 页

Exercise 71

解题思路

只保留被积函数为正的部分。

SOLUTION

For $a \le b$, $x - x^2 = x(1 - x)$ is positive on $(0, 1)$ and negative outside $[0, 1]$. Removing any exterior part cannot lower the integral, and including any missing portion of $(0, 1)$ increases it. Thus $a = 0$ and $b = 1$.

参考教材例题 3，第 289 页

Exercise 73

解题思路

找出区间上的最小值和最大值，再乘区间长度。

SOLUTION

On $[0, 1]$, $1/(1 + x^2)$ decreases from 1 to $1/2$. The Max-Min Inequality gives

$$\frac{1}{2} \le \int_0^1 \frac{dx}{1 + x^2} \le 1.$$

参考教材例题 3，第 289 页

Exercise 76

解题思路

根式递增，用两个端点值夹住积分。

SOLUTION

For $0 \le x \le 1$, $2\sqrt{2} \le \sqrt{x+8} \le 3$. The interval has length 1, so the integral lies between $2\sqrt{2}$ and 3.

参考教材例题 3，第 289 页

Exercise 77

解题思路

用零函数作逐点下界。

SOLUTION

Since 0 is a lower bound for f on $[a,b]$, the Max-Min Inequality yields $\int_a^b f(x)\,dx \ge 0(b-a) = 0$.

参考教材例题 3，第 289 页

Exercise 78

解题思路

把零作为上界，或对相反数用非负性的结论。

SOLUTION

The integrable function $-f$ is nonnegative. Thus $\int_a^b (-f(x))\,dx \ge 0$. By linearity, $\int_a^b f(x)\,dx \le 0$.

参考教材例题 5，第 292 页

Exercise 81

解题思路

把平均值视为常数，积分等于这个高度乘区间长度。

SOLUTION

Yes. For $a < b$,

$$\int_a^b \operatorname{av}(f)\,dx = (b-a)\operatorname{av}(f)$$

by the constant-function rule. The definition of $\operatorname{av}(f)$ makes the right side $\int_a^b f(x)\,dx$.

参考教材5.2 节例题 6，第 282 页；图 5.10，第 282 页

Exercise 83

解题思路

相邻端点的高度差相加会望远镜消去，不等宽时用最大宽度统一估计。

SOLUTION

On $[x_{k-1}, x_k]$, the height gap is $h_k = f(x_k) - f(x_{k-1}) \ge 0$.

(a) Each gap rectangle has width Δx. Translating the rectangles onto one common base stacks their disjoint height intervals, so they form a rectangle of height $\sum_k h_k = f(b) - f(a)$. Thus $U - L = \Delta x[f(b) - f(a)]$.

(b) Since $\Delta x_k \le \|P\|$,

$$0 \le U - L = \sum_k h_k \Delta x_k \le \|P\|[f(b) - f(a)].$$

The upper bound tends to zero as $\|P\| \to 0$.

5.4 The Fundamental Theorem of Calculus

21 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 305–307

参考教材例题 3，第 300 页

Exercise 1

解题思路

展开被积函数，再用原函数代入端点。

SOLUTION

By the Evaluation Theorem,

$$\int_0^2 x(x-3)\,dx = \left[x^3/3 - 3x^2/2\right]_0^2 = -10/3.$$

参考教材例题 3，第 300 页

Exercise 3

解题思路

把分母写成负幂，注意链式求导。

SOLUTION

An antiderivative is $-(x+3)^{-3}$. Its increment from -2 to 2 is $1 - 1/125 = 124/125$.

参考教材例题 3，第 300 页

Exercise 4

解题思路

根号 99 是常数，乘以奇函数后在对称区间抵消。

SOLUTION

The integrand is $\sqrt{99}\,x$, with antiderivative $\sqrt{99}\,x^2/2$. Its values at 1 and -1 agree, so the integral is 0.

参考教材例题 3，第 300 页

Exercise 8

解题思路

用幂函数的原函数，利用上限是 2 的五次方。

SOLUTION

An antiderivative of $x^{-6/5}$ is $-5x^{-1/5}$. Hence the integral is $-5(32^{-1/5}) + 5 = 5/2$.

参考教材例题 3，第 300 页

Exercise 9

解题思路

识别正切的导数。

SOLUTION

Since $(2\tan x)' = 2\sec^2 x$, the integral is $2\tan(\pi/3) - 2\tan 0 = 2\sqrt{3}$.

参考教材例题 3，第 300 页

Exercise 10

解题思路

分别积分常数项和余弦项。

SOLUTION

An antiderivative is $x + \sin x$. Its increment is $(\pi + \sin\pi) - (0 + \sin 0) = \pi$.

参考教材例题 3，第 300 页

Exercise 12

解题思路

把分式认成正割与正切的乘积。

SOLUTION

The integrand is $4\sec u\tan u = (4\sec u)'$. The integral is $4(\sec(\pi/3) - \sec 0) = 4$.

参考教材例题 3，第 300 页

Exercise 17

解题思路

找原函数时补上内层导数的系数。

SOLUTION

An antiderivative is $-\frac{1}{2}\cos(2x)$. The integral is $\frac{1}{2}(1-\cos(\pi/4)) = (2-\sqrt{2})/4$.

参考教材例题 3，第 300 页

Exercise 22

解题思路

逐项除以分母，化成幂函数之差。

SOLUTION

The integrand is $y^2 - 2y^{-2}$, with antiderivative $y^3/3 + 2/y$. Evaluating from -3 to -1 gives $22/3$.

参考教材例题 3，第 300 页

Exercise 24

解题思路

把乘积展开并除以分母，逐项积分。

SOLUTION

The integrand simplifies to $2 + 2x^{-1/3} - x^{1/3} - x^{2/3}$. An antiderivative is

$$F(x) = 2x + 3x^{2/3} - \frac{3}{4}x^{4/3} - \frac{3}{5}x^{5/3}.$$

Thus $F(8) - F(1) = -137/20$.

参考教材例题 8，第 304 页

Exercise 27

解题思路

在零点拆开绝对值，也可利用偶对称。

SOLUTION

Since $|x|$ is even and equals x for $x \geq 0$,

$$\int_{-4}^{4} |x|\,dx = 2\int_{0}^{4} x\,dx = 16.$$

参考教材例题 8，第 304 页

Exercise 28

解题思路

被积式只保留余弦为正的那一段。

SOLUTION

On $[0, \pi/2]$ the integrand is $\cos x$. On $[\pi/2, \pi]$ it is 0. Thus the integral is $\sin(\pi/2) - \sin 0 = 1$.

参考教材例题 3，第 300 页

Exercise 30

解题思路

从余弦复合函数出发猜原函数，再求导验证。

SOLUTION

Let $F(x) = -2\cos\sqrt{x}$. The chain rule gives $F'(x) = \sin\sqrt{x}/\sqrt{x}$. Therefore $F(\pi^2) - F(1) = 2 + 2\cos 1$.

参考教材例题 2，第 298 页

Exercise 40

解题思路

上限就是自变量，直接用微积分基本定理第一部分。

SOLUTION

Since $1/t$ is continuous on $(0, \infty)$, the Fundamental Theorem gives $dy/dx = 1/x$ for $x > 0$.

参考教材例题 2，第 298 页

Exercise 44

解题思路

先对外层立方求导，再对积分求导。

SOLUTION

Write $G(x) = \int_0^x (t^3+1)^{10}\,dt$. Then $G'(x) = (x^3+1)^{10}$ and $y' = 3G(x)^2(x^3+1)^{10}$.

参考教材例题 2，第 298 页

Exercise 46

解题思路

交换上下限带来负号，再用链式法则处理正切。

SOLUTION

The Fundamental Theorem and chain rule give

$$y' = -\frac{\sec^2 x}{1+\tan^2 x} = -1$$

wherever $\tan x$ is defined.

参考教材例题 7，第 303 页

Exercise 61

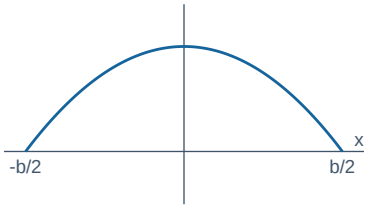
解题思路

抛物线关于纵轴对称，两端在横轴上，用偶对称积分。

SOLUTION

The arch opens downward with vertex $(0, h)$ and zeros at $\pm b/2$. Since it is nonnegative there,

$$A = 2\left[hx - \frac{4h}{3b^2}x^3\right]_0^{b/2} = \frac{2}{3}bh.$$



参考教材例题 4，第 301 页

Exercise 63

解题思路

边际成本是总成本的导数，对它积分得到成本增量。

SOLUTION

By the Net Change Theorem,

$$c(100) - c(1) = \int_1^{100} \frac{dx}{2\sqrt{x}} = \left[\sqrt{x}\right]_1^{100} = 9$$

dollars.

参考教材例题 2，第 298 页

Exercise 68

解题思路

等式两边求导，用乘积法则后代入 4。

SOLUTION

Differentiating gives $f(x) = \cos(\pi x) - \pi x \sin(\pi x)$. Hence $f(4) = 1$.

参考教材例题 2，第 298 页

Exercise 71

解题思路

先把积分函数的一、二阶导数写成原函数及其导数，再判定符号。

SOLUTION

The Fundamental Theorem gives $g' = f$ and $g'' = f' > 0$. Hence (a) and (b) hold. Since $f(1) = 0$, (c) holds. Since f is strictly increasing, g' changes from negative to positive at 1. Thus (e) and (g) hold, and (d) fails. The graph is concave up throughout, so (f) fails.

参考教材例题 5，第 302 页

Exercise 73

解题思路

速度读曲线高度，加速度读斜率，位置读带符号的累计面积。

SOLUTION

Here $v = f$ and $a = f'$.

(a) $v(5) = 2$ m/s.

(b) $a(5) < 0$, since the graph is falling.

(c) The graph is the line $y = x$ on $[0, 3]$, so $s(3) = 9/2$ m.

(d) s is maximal at $t = 6$, where v changes from positive to negative.

(e) Horizontal tangents give $a = 0$ near $t = 4$ and $t = 7$.

(f) The particle moves away on $(0, 6)$ and toward the origin on $(6, 9)$.

(g) It lies on the positive side. The positive area already exceeds $9/2 + 4$ by $t = 5$, while the negative area has magnitude less than $3(2) = 6$. Thus s stays positive for $0 < t \leq 9$.

5.5 Indefinite Integrals and the Substitution Method

12 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 313–314

参考教材例题 2，第 309 页

Exercise 18

解题思路

把根号内的线性式作为新变量。

SOLUTION

Set $u = 5s + 4$, so $du = 5\,ds$. The integral is $\frac{1}{5}\int u^{-1/2}\,du = \frac{2}{5}\sqrt{5s+4} + C$.

参考教材例题 1，第 308 页

Exercise 20

解题思路

根式内的二次式求导后，与前面的因子只差常数。

SOLUTION

With $u = 7 - 3y^2$, $du = -6y\,dy$. The integral becomes $-\frac{1}{2}\int u^{1/2}\,du = -\frac{1}{3}(7-3y^2)^{3/2} + C$.

参考教材例题 2，第 309 页

Exercise 21

解题思路

令新变量为一加根号，分母中的根号正好出现在微分里。

SOLUTION

Set $u = 1 + \sqrt{x}$, so $dx/\sqrt{x} = 2\,du$. The result is $2\int u^{-2}\,du = -2/(1+\sqrt{x}) + C$.

参考教材例题 7，第 311 页

Exercise 22

解题思路

把余弦立方拆成一个余弦和一项平方，再用正弦换元。

SOLUTION

Use $u = \sin x$, $du = \cos x\,dx$, and $\cos^2 x = 1 - u^2$. Then

$$\int u^{1/2}(1-u^2)\,du = \frac{2}{3}u^{3/2} - \frac{2}{7}u^{7/2} + C.$$

Substitute $u = \sin x$.

参考教材例题 3，第 310 页

Exercise 26

解题思路

正割平方提示取正切为新变量。

SOLUTION

For $u = \tan(x/2)$, $du = \frac{1}{2}\sec^2(x/2)\,dx$. Thus the integral is $2\int u^7\,du = \frac{1}{4}\tan^8(x/2) + C$.

参考教材例题 4，第 310 页

Exercise 31

解题思路

取分母中的余弦换元，分子的正弦提供微分。

SOLUTION

Set $u = \cos(2t+1)$, so $du = -2\sin(2t+1)\,dt$. The result is $-\frac{1}{2}\int u^{-2}\,du = \frac{1}{2}\sec(2t+1) + C$.

参考教材例题 8，第 312 页

Exercise 38

解题思路

先从根号内提出完全平方，剩下的式子适合换元。

SOLUTION

Rewrite the integrand as $x^{-2}\sqrt{1-1/x}$. With $u = 1 - 1/x$, $du = x^{-2}\,dx$. Thus the result is $\frac{2}{3}(1-1/x)^{3/2} + C$ on each real interval of definition.

参考教材例题 8，第 312 页

Exercise 42

解题思路

把分子根式化成平方项，再取分母里的三次式。

SOLUTION

On the real domain $x > 1$, the integrand is $x^2/\sqrt{x^3-1}$. Set $u = x^3 - 1$ to obtain $\frac{1}{3}\int u^{-1/2}\,du = \frac{2}{3}\sqrt{x^3-1} + C$.

参考教材例题 6，第 311 页

Exercise 43

解题思路

取括号内为新变量，再把外面的横坐标也改写。

SOLUTION

Let $u = x - 1$. Then $x = u + 1$ and the integral is

$$\int (u^{11} + u^{10})\,du = \frac{(x-1)^{12}}{12} + \frac{(x-1)^{11}}{11} + C.$$

参考教材例题 6，第 311 页

Exercise 46

解题思路

换元后将线性因子拆开，再逐项积分。

SOLUTION

Set $u = x - 5$. Then $x + 5 = u + 10$ and the integral is

$$\int (u^{4/3} + 10u^{1/3})\,du = \tfrac{3}{7}u^{7/3} + \tfrac{15}{2}u^{4/3} + C.$$

Replace u by $x - 5$.

参考教材例题 1，第 308 页

Exercise 56

解题思路

先换元求原函数，再代入初值确定常数。

SOLUTION

With $u = x^2 + 8$, $dy = 2u^{-1/3}\,du$. Hence $y = 3(x^2 + 8)^{2/3} + C$. The condition $y(0) = 0$ gives $C = -12$.

参考教材例题 4，第 310 页

Exercise 62

解题思路

从加速度积分两次，每次用一个初值。

SOLUTION

Integrating $a(t)$ gives $v(t) = \pi\sin(\pi t) + C_1$. Since $v(0) = 8$, $C_1 = 8$. Integrating again gives $s(t) = -\cos(\pi t) + 8t + C_2$. Since $s(0) = 0$, $C_2 = 1$. Thus $s(1) = 10$ m.

5.6 Definite Integral Substitutions and the Area Between Curves

17 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 321–324

参考教材例题 2，第 315 页

Exercise 8

解题思路

取分母里的表达式作新变量，分别换算两组上下限。

SOLUTION

Let $u = 1 + v^{3/2}$, so $10\sqrt{v}\,dv = \frac{20}{3}du$. The transformed antiderivative is $-20/(3u)$.

(a) The bounds become $1, 2$, giving $10/3$.

(b) They become $2, 9$, giving $70/27$.

参考教材例题 2，第 315 页

Exercise 11

解题思路

取根号内为新变量，把外面的线性因子也一起改写。

SOLUTION

Let $u = 4 + 5t$. Then $t\sqrt{4+5t}\,dt = (u-4)\sqrt{u}\,du/25$. An antiderivative in u is $G(u) = 2u^{5/2}/125 - 8u^{3/2}/75$.

(a) $G(9) - G(4) = 506/375$.

(b) $G(49) - G(9) = 86744/375$.

参考教材例题 2，第 315 页

Exercise 12

解题思路

括号求导后恰好得到正弦因子乘常数。

SOLUTION

Set $u = 1 - \cos 3t$, so $du = 3\sin 3t\,dt$. The original bounds $0, \pi/6, \pi/3$ map to $0, 1, 2$. Evaluate $u^2/6$ over the two intervals to obtain (a) $1/6$ and (b) $1/2$.

参考教材例题 2，第 315 页

Exercise 21

解题思路

被积式是复合幂函数乘以内层的导数。

SOLUTION

Set $u = 4y - y^2 + 4y^3 + 1$. Its differential is $(12y^2 - 2y + 4)\,dy$. Hence the integral is $\int_1^8 u^{-2/3}\,du = 3(2 - 1) = 3$.

参考教材例题 2，第 315 页

Exercise 23

解题思路

换元后得到一个周期内的余弦平方积分。

SOLUTION

Set $u = \theta^{3/2}$, so $\sqrt{\theta}\,d\theta = \frac{2}{3}du$. The upper bound becomes π. Using $\cos^2 u = (1 + \cos 2u)/2$ gives

$$\frac{2}{3}\int_0^{\pi} \cos^2 u\,du = \frac{\pi}{3}.$$

参考教材例题 3，第 317 页；5.4 节例题 8，第 304 页

Exercise 25

解题思路

两块阴影关于原点对称，取右半面积再乘二。

SOLUTION

The integrand is odd, with positive sign on $(0, 2)$. The total shaded area is

$$2\int_0^2 x\sqrt{4 - x^2}\,dx = \int_0^4 \sqrt{u}\,du = \frac{16}{3},$$

using $u = 4 - x^2$.

参考教材例题 4，第 318 页

Exercise 59

解题思路

先联立求交点，再用上方抛物线减下方四次曲线。

SOLUTION

The curves are $y = 4 - 4x^2$ and $y = x^4 - 1$. Their intersections satisfy $x^4 + 4x^2 - 5 = 0$, giving $x = \pm 1$. The parabola is upper between them, so

$$A = 2\int_0^1 (5 - 4x^2 - x^4)\,dx = \frac{104}{15}.$$

参考教材例题 4，第 318 页

Exercise 63

解题思路

用二倍角公式比较两条曲线的高低。

SOLUTION

On $[0, \pi]$, the difference $2\sin x - \sin 2x = 2\sin x(1 - \cos x)$ is nonnegative. Hence

$$A = \left[-2\cos x + \frac{1}{2}\cos 2x\right]_0^{\pi} = 4.$$

参考教材例题 6，第 319 页

Exercise 71

解题思路

以水平切片量宽度，利用两块区域的中心对称。

SOLUTION

The curves are $x = y^3$ and $x = y$, meeting at $y = -1, 0, 1$. On $[0, 1]$ the line is to the right. Thus

$$A = 2\int_0^1 (y - y^3)\,dy = \frac{1}{2}.$$

参考教材例题 6，第 319 页

Exercise 72

解题思路

把两个根式关系改写成以横坐标为自变量的幂函数。

SOLUTION

The curves are $y = x^3$ and $y = x^5$, intersecting at $x = -1, 0, 1$. By central symmetry the area is

$$2\int_0^1 (x^3 - x^5)\,dx = \frac{1}{6}.$$

参考教材例题 6，第 319 页

Exercise 78

解题思路

水平切片在高度 1 处同时换左右边界，到高度 2 收口。

SOLUTION

For $0 \le y \le 1$, the horizontal width is $2\sqrt{y}$. For $1 \le y \le 2$, it is $3 - y - (y - 1)^2$. The change occurs where $2\sqrt{y} = 3 - y$ and the left boundary leaves the axis. The second strip ends at $(1, 2)$.

The two areas are $4/3$ and $7/6$, so $A = 5/2$.

参考教材例题 4–5，第 318 页

Exercise 82

解题思路

真实面积累计竖直距离，比较它与有符号差的积分。

SOLUTION

Sometimes true. The area is $\int_a^b |f - g|\,dx$. It equals $\int_a^b (f - g)\,dx$ exactly when $f \ge g$ throughout $[a, b]$. Indeed, if $f < g$ anywhere, continuity gives a subinterval with strictly positive difference between the two integrands.

参考教材例题 2，第 315 页

Exercise 83

解题思路

把正弦的自变量换为新变量，原函数可直接使用。

SOLUTION

With $u = 2x$, $dx/x = du/u$. Therefore the integral is $\int_2^6 (\sin u)/u \, du = F(6) - F(2)$.

参考教材例题 2，第 315 页

Exercise 84

解题思路

把区间左右反射，微分的负号与上下限交换抵消。

SOLUTION

Set $u = 1 - x$. Then

$$\int_0^1 f(1-x)\,dx = -\int_1^0 f(u)\,du = \int_0^1 f(u)\,du.$$

参考教材例题 3，第 317 页

Exercise 85

解题思路

用相反数把左半区间映到右半区间，再看奇偶性。

SOLUTION

Substituting $u = -x$ gives $\int_{-1}^0 f(x)\,dx = \int_0^1 f(-u)\,du$. It equals (a) -3 if f is odd and (b) 3 if f is even.

参考教材例题 3，第 317 页

Exercise 86

解题思路

奇函数在对称区间两侧的积分符号相反。

SOLUTION

For (a), substitution gives $\int_{-a}^0 f(x)\,dx = \int_0^a f(-u)\,du = -\int_0^a f(u)\,du$. Adding yields 0.

For (b), $\int_{-\pi/2}^{\pi/2} \sin x \, dx = [-\cos x]_{-\pi/2}^{\pi/2} = 0$.

参考教材例题 2，第 315 页

Exercise 87

解题思路

反射后两个分子的和等于公共分母。

SOLUTION

Where the displayed integrand is defined and integrable, substituting $u = a - x$ gives

$$I = \int_0^a \frac{f(a-x)}{f(x) + f(a-x)}\,dx.$$

Adding this to the original formula gives $2I = \int_0^a 1\,dx = a$, so $I = a/2$. For example, this applies if the denominator is nowhere zero on the interval.

6.1 Volumes Using CrossSections

13 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 339–343

参考教材例题 1，第 333 页

Exercise 1

解题思路

先求正方形的对角线，再由对角线求截面积。

SOLUTION

At x, the diagonal is $2\sqrt{2x}$, so the square has area $(2\sqrt{2x})^2/2 = 4x$. Hence $V = \int_0^4 4x\,dx = 32$.

参考教材例题 2，第 334 页

Exercise 10

解题思路

圆盘内的水平弦是等腰直角三角形的一条直角边。

SOLUTION

At height y, the chord has length $2\sqrt{1-y^2}$. The cross-sectional area is half its square, $2(1-y^2)$. Thus $V = \int_{-1}^{1} 2(1-y^2)\,dy = 8/3$.

参考教材例题 1，第 333 页

Exercise 12

解题思路

用相似比求距顶点给定高度处的正方形边长。

SOLUTION

At distance u from the apex, the side is $3u/5$ by similarity. The volume is $\int_0^5 (3u/5)^2\,du = 15$.

参考教材例题 3，第 334 页

Exercise 14

解题思路

在同一高度比较两立体的圆截面。

SOLUTION

The disk at x has diameter $x - x/2 = x/2$, hence radius $x/4$. A cone with apex at $x = 0$, height 12, and base radius 3 has radius $3x/12 = x/4$ at the same x. The matching cross sections give equal volumes by Cavalieri's principle, namely 36π.

参考教材例题 4，第 335 页

Exercise 21

解题思路

绕横轴旋转时，竖直切片产生半径为曲线高度的圆盘。

SOLUTION

The disk radius is x^2 on $[0,2]$. Thus $V = \pi\int_0^2 x^4\,dx = 32\pi/5$.

参考教材例题 4，第 335 页

Exercise 25

解题思路

圆盘面积是半径的平方乘圆周率，根式因此消去。

SOLUTION

The disk radius is $\sqrt{\cos x}$. Hence $V = \pi\int_0^{\pi/2} \cos x\,dx = \pi$.

参考教材例题 7，第 336 页

Exercise 29

解题思路

绕纵轴转，用水平切片和横向半径。

SOLUTION

The radius is $\sqrt{5}y^2$, with $-1 \le y \le 1$. Thus $V = 5\pi \int_{-1}^{1} y^4\,dy = 2\pi$.

参考教材例题 7，第 336 页

Exercise 31

解题思路

用横坐标作圆盘半径，平方后直接积分。

SOLUTION

The squared radius is $2\sin 2y$ on $[0, \pi/2]$. Thus $V = \pi\left[-\cos 2y\right]_0^{\pi/2} = 2\pi$.

参考教材例题 9，第 338 页

Exercise 39

解题思路

先解交点，垫圈面积用外半径平方减内半径平方。

SOLUTION

The intersections are $x = -1, 2$. The outer radius is $x + 3$ and the inner radius is $x^2 + 1$. Hence

$$V = \pi \int_{-1}^{2} (-x^4 - x^2 + 6x + 8)\,dx = \frac{117\pi}{5}.$$

参考教材例题 10，第 339 页

Exercise 43

解题思路

水平切片从直线横坐标 1 到斜边横坐标。

SOLUTION

At $y \in [0, 1]$, the inner radius is 1 and the outer radius is $1 + y$. Thus

$$V = \pi \int_0^1 ((1+y)^2 - 1)\,dy = \frac{4\pi}{3}.$$

参考教材例题 10，第 339 页

Exercise 45

解题思路

把抛物线改写为横坐标，用水平垫圈。

SOLUTION

For $0 \le y \le 4$, the horizontal strip runs from $x = \sqrt{y}$ to $x = 2$. Its washer area is $\pi(4 - y)$, giving $V = 8\pi$.

参考教材例题 10，第 339 页

Exercise 53

解题思路

水平切片绕轴产生垫圈，展开两半径的平方差。

SOLUTION

Write $r(y) = \sqrt{a^2 - y^2}$. The outer and inner radii about $x = b$ are $b + r$ and $b - r$. Their squared difference is $4br$. Hence

$$V = 4\pi b \int_{-a}^{a} r(y)\,dy = 2\pi^2 a^2 b,$$

using the area of a semicircle.

参考教材例题 9，第 338 页

Exercise 61

解题思路

两次体积相减可消去二次项。绕下方直线旋转时，切片中央留有空心。

SOLUTION

About $y = 0$ the cross-sectional area is πf^2. About $y = -1$ the inner radius is 1 and the outer radius is $f + 1$. Thus

$$8\pi - 4\pi = 2\pi \int_a^b f(x)\,dx,$$

and the area is 2. The printed key retains a term involving $b - a$. That term cancels when the hole of radius 1 is included in the washer formula.

6.3 Arc Length

5 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 355–356

参考教材例题 2，第 352 页

Exercise 1

解题思路

求导后检查弧长根式是否为完全平方。

SOLUTION

Here $y' = x\sqrt{x^2+2}$, so $1+(y')^2 = (x^2+1)^2$. Thus $L = \int_0^3 (x^2+1)\,dx = 12$.

参考教材例题 1，第 352 页

Exercise 2

解题思路

弧长根号内为一次式，可以直接换元积分。

SOLUTION

Since $y' = \frac{3}{2}\sqrt{x}$,

$$L = \int_0^4 \sqrt{1+9x/4}\,dx = \frac{8}{27}(10\sqrt{10}-1).$$

参考教材例题 2，第 352 页

Exercise 9

解题思路

把导数中的减号改为加号，就得到弧长根式的正平方根。

SOLUTION

Here $y' = x^2 - 1/(4x^2)$ and $1+(y')^2 = (x^2+1/(4x^2))^2$. Hence

$$L = \left[x^3/3 - 1/(4x)\right]_1^3 = \frac{53}{6}.$$

参考教材例题 1，第 352 页；5.4 节例题 2，第 298 页

Exercise 23

解题思路

先对积分函数求导，再用余弦二倍角化简。

SOLUTION

The Fundamental Theorem gives $y' = \sqrt{\cos 2x}$. Thus the speed factor is $\sqrt{1+\cos 2x} = \sqrt{2}\cos x$ on $[0, \pi/4]$, giving $L = \sqrt{2}\sin(\pi/4) = 1$.

参考教材弧长公式的推导，第 351–352 页

Exercise 30

解题思路

切线段的横向和纵向变化组成直角三角形，长度之和是黎曼和。

SOLUTION

For (a), the run is Δx_k and the rise is $f'(x_{k-1})\Delta x_k$. The Pythagorean theorem gives length $\Delta x_k\sqrt{1+f'(x_{k-1})^2}$.

For (b), these lengths form a left-endpoint Riemann sum for the continuous function $\sqrt{1+(f')^2}$. As the mesh tends to zero, the sum tends to $\int_a^b \sqrt{1+(f'(x))^2}\,dx$.

6.4 Areas of Surfaces of Revolution

6 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 360–361

参考教材例题 1，第 358 页

Exercise 13

解题思路

旋转半径是纵坐标。弧长根号内的导数可配合外面的三次项换元。

SOLUTION

With $y' = x^2/3$,

$$S = \frac{2\pi}{9}\int_0^2 x^3\sqrt{1 + x^4/9}\,dx = \frac{98\pi}{81},$$

by substituting $u = 1 + x^4/9$.

参考教材例题 2，第 359 页

Exercise 18

解题思路

曲线在纵轴左侧，旋转半径取横坐标的绝对值。

SOLUTION

Here $x' = (y-1)/(2\sqrt{y})$, so $ds/dy = (y+1)/(2\sqrt{y})$. Since $x \le 0$, the radius is $\sqrt{y}(1 - y/3)$. Thus

$$S = \pi\int_1^3 (1 - y/3)(y+1)\,dy = 16\pi/9.$$

参考教材例题 2，第 359 页

Exercise 22

解题思路

绕纵轴的半径为横坐标，但弧长仍可用横坐标积分。

SOLUTION

Since $y' = x\sqrt{x^2+2}$, $ds = (1+x^2)\,dx$. Therefore

$$S = 2\pi\int_0^{\sqrt{2}} x(1+x^2)\,dx = 4\pi.$$

参考教材例题 1，第 358 页

Exercise 23

解题思路

半径为纵坐标，用纵坐标表达弧长元素。

SOLUTION

Here $dx/dy = y^3 - 1/(4y^3)$, giving $ds/dy = y^3 + 1/(4y^3)$. Then

$$S = 2\pi\left[y^5/5 - 1/(4y)\right]_1^2 = \frac{253\pi}{20}.$$

参考教材例题 1，第 358 页

Exercise 25

解题思路

半圆的旋转半径与弧长根式中的分母正好抵消。

SOLUTION

For $-a < x < a$, $y' = -x/\sqrt{a^2 - x^2}$, so $y\sqrt{1 + (y')^2} = a$. The integral extends to the endpoints by limits, giving $S = 2\pi a\int_{-a}^{a} dx = 4\pi a^2$.

参考教材例题 1，第 358 页

Exercise 28

解题思路

球带与整球使用同一个常数面积密度，只改变积分区间。

SOLUTION

For the semicircle, $y\sqrt{1 + (y')^2} = r$ at every interior point. Thus the area over any interval $[a, a+h]$ is $\int_a^{a+h} 2\pi r\,dx = 2\pi r h$, independent of a.

6.5 Work and Fluid Forces

9 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 367–371

参考教材例题 2，第 363 页

Exercise 3

解题思路

用胡克定律表示拉力，对伸长量积分并反求劲度系数。

SOLUTION

The extension ranges from 0 to 3 m. Hooke's law gives $1800 = \int_0^3 kx\,dx = 9k/2$, so $k = 400$ N/m.

参考教材例题 4，第 364 页

Exercise 9

解题思路

离顶端越远的绳段需要提升越多。

SOLUTION

The element initially x meters below the top weighs $0.624\,dx$ N and rises x m. Thus $W = 0.624\int_0^{50} x\,dx = 780$ J.

参考教材例题 1，第 362 页

Exercise 12

解题思路

把朝向原点的力写成负分量，同时积分方向也是从右到左。

SOLUTION

For $x > 0$, $F_x = -k/x^2$. Therefore

$$W = \int_b^a (-k/x^2)\,dx = k(1/a - 1/b).$$

参考教材例题 5，第 364 页

Exercise 16

解题思路

用距地面的深度积分，抽到一半时先抽走上层的水。

SOLUTION

Let z be depth below ground, from 3 to 6 m. Each slice has volume $24\,dz$. If water has weight density γ,

$$W = 24\gamma\int_3^6 z\,dz = 324\gamma.$$

The upper half requires $24\gamma\int_3^{4.5} z\,dz = 135\gamma$.

(a) For $\gamma = 9800$, $W = 3175200$ J.

(b) At 370 W the time is $W/370 \approx 143.0$ min.

(c) Half the water takes $135\gamma/370 \approx 59.6$ min.

(d) For $\gamma = 9780$, the results are 3168720 J, 142.7 min and 59.5 min. For $\gamma = 9820$, they are 3181680 J, 143.3 min and 59.7 min.

参考教材功的积分公式，第 363 页

Exercise 25

解题思路

用链式法则把加速度改写成速度对位置的导数。

SOLUTION

With signed force and velocity, $F(x) = m\,dv/dt = mv\,dv/dx$ along the motion. Hence

$$\int_{x_1}^{x_2} F(x)\,dx = \left[\tfrac{1}{2}mv(x)^2\right]_{x_1}^{x_2}.$$

Equivalently, $F\,dx/dt = d(mv^2/2)/dt$, which gives the same result even at an instant of rest.

参考教材本节第 25 题，第 369 页

Exercise 26

解题思路

把克换算为千克，再算从静止到发球速度的动能增量。

SOLUTION

By the work-energy identity, $W=\frac{1}{2}(0.060)(50)^2=75$ J.

参考教材例题 5，第 364 页

Exercise 31

解题思路

用相似关系表示圆台截面半径，每层都提升到吸管顶端。

SOLUTION

At height y cm above the base, the radius is $3+y/12$ cm and the lift is $21-y$ cm. The weight density is 7840 N/m^3. Converting cubic centimeters and centimeters to SI units gives

$$W=\frac{7840\pi}{10^8}\int_0^{18}(3+y/12)^2(21-y)\,dy$$

or approximately 0.863 J.

参考教材例题 6，第 366 页

Exercise 38

解题思路

每条水平薄片所受压力等于深度乘重度，再乘薄片面积。

SOLUTION

At height y above the bottom the width is $2\sqrt{25-y^2}$ and the depth is $6-y$. Thus

$$F=9800\int_0^5 2(6-y)\sqrt{25-y^2}\,dy.$$

The first term uses the quarter-circle area, and the y term uses $u=25-y^2$. This gives $9800(75\pi-250/3)$ N.

参考教材例题 6，第 366 页

Exercise 46

解题思路

先由端板承压求最大水深，再把端面面积乘槽长。

SOLUTION

At height y the width is $2\sqrt{y/3}$. For depth $h\le 3$,

$$F(h)=9800\int_0^h 2(h-y)\sqrt{y/3}\,dy=\frac{78400}{15\sqrt{3}}h^{5/2}.$$

Solving $F=25000$ gives $h\approx 2.330$ m. Then

$$V=10\int_0^h 2\sqrt{y/3}\,dy=\frac{40}{3\sqrt{3}}h^{3/2}\approx 27.374$$

m^3. The safe whole-number capacity is 27 m^3, corresponding to a depth of about 2.308 m.

7.1 Inverse Functions and Their Derivatives

16 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 394–396

参考教材例题 1，第 389 页

Exercise 1

解题思路

用水平线检验，严格单调函数自动通过。

SOLUTION

Yes. The graph of $-3x^3$ is strictly decreasing, so every horizontal line meets it at most once.

参考教材例题 1，第 389 页

Exercise 2

解题思路

寻找同一高度上不同的横坐标。

SOLUTION

No. The horizontal line $y = 0$ intersects $y = x^4 - x^2$ at $x = -1, 0, 1$.

参考教材例题 1，第 389 页

Exercise 3

解题思路

绝对值图形左右对称，取一条横线即可否定。

SOLUTION

No. For instance, the horizontal line $y = 2$ intersects $y = 2|x|$ at $x = -1$ and $x = 1$.

参考教材例题 1，第 389 页

Exercise 4

解题思路

阶梯图上的一整段有相同函数值。

SOLUTION

No. The graph is constant on each interval $[k, k+1)$, so a horizontal line at a step meets the graph more than once.

参考教材例题 1，第 389 页

Exercise 5

解题思路

用倒数的代数性质检查两个分支之间是否重复取值。

SOLUTION

Yes. For nonzero x_1, x_2, the equation $1/x_1 = 1/x_2$ implies $x_1 = x_2$.

参考教材例题 1，第 389 页

Exercise 6

解题思路

立方根在全实轴严格递增。

SOLUTION

Yes. The cube root is strictly increasing on $\mathbb{R}$, so every horizontal line meets its graph once.

参考教材例题 3，第 391 页；图 7.5，第 390 页

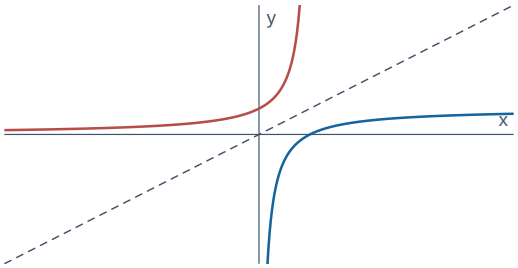
Exercise 12

解题思路

把原曲线沿直线 $y = x$ 反射，定义域和值域互换。

SOLUTION

Solving $y = 1 - 1/x$ for $x > 0$ gives $x = 1/(1-y)$. Thus $f^{-1}(x) = 1/(1-x)$ with domain $(-\infty, 1)$ and range $(0, \infty)$. Reflecting the original graph interchanges its asymptotes with $x = 1$ and $y = 0$ for the inverse.



Blue: f Red: inverse Dashed: y = x

参考教材图 7.5，第 390 页

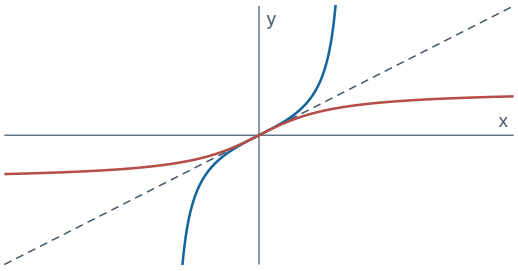
Exercise 14

解题思路

反射正切的指定分支，原来的竖直渐近线变成水平渐近线。

SOLUTION

The reflection is $y = \arctan x$, with domain $\mathbb{R}$ and range $(-\pi/2, \pi/2)$. Its horizontal asymptotes are $y = \pm\pi/2$.



Blue: tangent on its principal branch Red: inverse

参考教材例题 4，第 391 页

Exercise 17

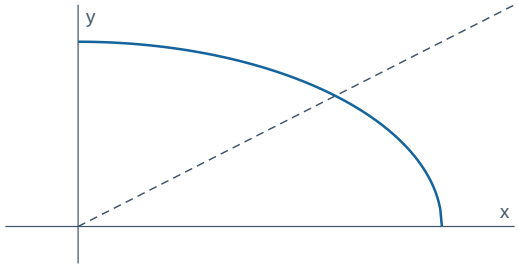
解题思路

图形是第一象限的四分之一圆，交换坐标后仍在同一圆弧上。

SOLUTION

For (a), $x^2 + y^2 = 1$ with $x, y \geq 0$, so the arc is symmetric about $y = x$.

For (b), $f(f(x)) = \sqrt{1 - (1 - x^2)} = x$ on $[0, 1]$. Thus $f^{-1} = f$.



The quarter-circle is invariant under reflection.

参考教材例题 3，第 391 页

Exercise 25

解题思路

交换输入与输出，再取实数五次根。

SOLUTION

The inverse is $f^{-1}(x) = x^{1/5}$. Its domain and range are both $\mathbb{R}$. The identities $(x^{1/5})^5 = x$ and $(x^5)^{1/5} = x$ verify both compositions.

参考教材例题 3，第 391 页

Exercise 31

解题思路

解分式方程，分母不能为零决定排除点。

SOLUTION

Solving $y = (x+3)/(x-2)$ for x gives $x = (2y+3)/(y-1)$. Thus $f^{-1}(x) = (2x+3)/(x-1)$, with domain $\mathbb{R} \setminus \{1\}$ and range $\mathbb{R} \setminus \{2\}$. Substitution gives both identity compositions on their respective domains.

参考教材例题 5–6，第 393 页

Exercise 38

解题思路

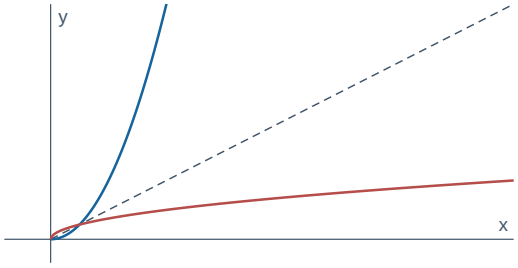
先求非负分支的反函数，再比较对应点的导数。

SOLUTION

For (a), $f^{-1}(x) = \sqrt{x/2}$ on $[0, \infty)$.

(b) The graphs are reflections in $y = x$.

(c) $f(5) = 50$ and $f'(5) = 20$. Also $(f^{-1})'(x) = 1/(2\sqrt{2x})$ for $x > 0$, so $(f^{-1})'(50) = 1/20$.



Blue: f Red: inverse Dashed: y = x

参考教材例题 4–5，第 391、393 页

Exercise 39

解题思路

互为反函数的曲线关于对角线对称，在对应点的非零斜率互为倒数。

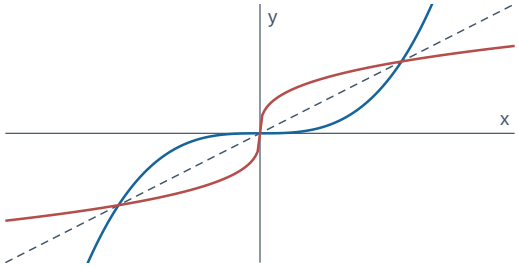
SOLUTION

For (a), cubing and taking the real cube root are inverse operations.

(b) The graphs are reflections across $y = x$ and meet at $(-1,-1), (0,0), (1,1)$.

(c) At each of $x = \pm 1$, the slopes are $f' = 3$ and $g' = 1/3$.

(d) At the origin, f has tangent $y = 0$ and g has tangent $x = 0$ by reflection.



Blue: cube Red: real cube root Dashed: y = x

参考教材例题 6，第 393 页

Exercise 44

解题思路

原函数在原点的斜率非零，直接取倒数。

SOLUTION

Because $g(0) = 0$ and $g'(0) = 2$, the inverse derivative theorem gives $(g^{-1})'(0) = 1/2$.

参考教材例题 1，第 389 页

Exercise 56

解题思路

若倒数相等且分母非零，则原来的值相等。

SOLUTION

Yes. If $h(x_1) = h(x_2)$, then $f(x_1) = f(x_2)$ because neither value is zero. Injectivity of f gives $x_1 = x_2$.

参考教材例题 6，第 393 页

Exercise 59

解题思路

对复合恒等式求导，得出导数的乘积关系。

SOLUTION

Differentiating $g(f(x)) = x$ gives $g'(f(x))f'(x) = 1$. In particular $f'(x) \neq 0$ and $g'(f(x)) = 1/f'(x)$.

7.2 Natural Logarithms

18 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 403–404

参考教材例题 2，第 399 页

Exercise 1

解题思路

先把真数拆成 2 和 3 的幂，再用对数的乘除和幂法则。

SOLUTION

The factorizations are $3/4$, $4/9$, $1/2$, $3^{2/3}$, $3\,2^{1/2}$, and $3^{3/2}2^{-1/2}$. Their logarithms are, respectively:

(a) $\ln 3 - 2\ln 2$.

(b) $2\ln 2 - 2\ln 3$.

(c) $-\ln 2$.

(d) $\frac{2}{3}\ln 3$.

(e) $\ln 3 + \frac{1}{2}\ln 2$.

(f) $\frac{3}{2}\ln 3 - \frac{1}{2}\ln 2$.

参考教材例题 2，第 399 页

Exercise 4

解题思路

合并为一个对数，同时保留原式定义域。

SOLUTION

For (a), the arguments multiply to 1, giving 0 wherever $\cos u > 0$.

For (b), the quotient is $(8x+4)/4 = 2x+1$, giving $\ln(2x+1)$ for $x > -1/2$.

For (c), the quotient is $(t^2-1)/(t+1) = t-1$, giving $\ln(t-1)$ on the original domain $t > 1$.

参考教材例题 1，第 398 页

Exercise 7

解题思路

先把常数乘积拆开或直接用链式法则。

SOLUTION

For $x > 0$, $y' = 3/(3x) = 1/x$.

参考教材例题 1–2，第 398–399 页

Exercise 10

解题思路

把对数内的幂移到外面。

SOLUTION

On $t > 0$, $y = \frac{3}{2}\ln t$, so $dy/dt = 3/(2t)$.

参考教材例题 1–2，第 398–399 页

Exercise 11

解题思路

把商的对数拆成常数减去变量的对数。

SOLUTION

For $x > 0$, $y = \ln 3 - \ln x$, so $y' = -1/x$.

参考教材例题 1，第 398 页

Exercise 23

解题思路

使用商法则，分子里两项对数会抵消。

SOLUTION

The quotient rule gives $y' = 1/[x(1+\ln x)^2]$ for $x > 0$ and $x \neq e^{-1}$.

参考教材例题 1，第 398 页

Exercise 25

解题思路

内外两层对数分别求导。

SOLUTION

Since the outer argument must be positive, $x > 1$. The chain rule gives $y' = 1/(x \ln x)$.

参考教材例题 1–2，第 398–399 页

Exercise 36

解题思路

把根式和幂的指数移到对数外，再求导。

SOLUTION

For $x > -1$, $y = \frac{5}{2}\ln(x+1) - 10\ln(x+2)$. Thus $y' = 5/[2(x+1)] - 10/(x+2)$.

参考教材例题 3，第 401 页

Exercise 41

解题思路

分子是分母的导数，直接用对数积分。

SOLUTION

With $u = y^2 - 25$, $du = 2y\,dy$. The integral is $\ln|y^2 - 25| + C$ on each interval avoiding $y = \pm 5$.

参考教材例题 3，第 401 页

Exercise 44

解题思路

分子是分母的导数，绝对值要保留到代入端点。

SOLUTION

The integrand is the derivative of $\ln|1 - 4\cos u|$. Its denominator stays negative on $[0, \pi/3]$. The integral is $\ln 1 - \ln 3 = -\ln 3$.

参考教材例题 3，第 401 页

Exercise 47

解题思路

把自然对数作新变量，剩下的是负二次幂。

SOLUTION

Set $u = \ln x$. The integral is

$$\left[-1/u\right]_{\ln 2}^{\ln 4} = \frac{1}{\ln 2} - \frac{1}{\ln 4} = \frac{1}{2\ln 2}.$$

参考教材例题 4，第 402 页

Exercise 55

解题思路

先从分母提出根号，再令一加根号为新变量。

SOLUTION

Since $2\sqrt{x} + 2x = 2\sqrt{x}(1 + \sqrt{x})$, set $u = 1 + \sqrt{x}$. The integral becomes $\int du/u = \ln(1 + \sqrt{x}) + C$ for $x > 0$.

参考教材例题 5，第 402 页

Exercise 58

解题思路

根式含平方，先写成绝对值，再在两侧分别作对数求导。

SOLUTION

For $x \neq 1$, $y = |x - 1|\sqrt{x^2 + 1}$, so

$$y' = y\left(\frac{x}{x^2+1} + \frac{1}{x-1}\right).$$

The one-sided derivatives at 1 are $-\sqrt{2}$ and $\sqrt{2}$, so y' is undefined there.

参考教材例题 5，第 402 页

Exercise 67

解题思路

用绝对值的对数处理可能为负的因子，最后单独检查零点。

SOLUTION

For $x \neq -1, 0$, take $\ln|y| = \ln|x| + \frac{1}{2}\ln(x^2+1) - \frac{2}{3}\ln|x+1|$. Then

$$y' = y\left(\frac{1}{x} + \frac{x}{x^2+1} - \frac{2}{3(x+1)}\right).$$

At 0, the difference quotient gives $y'(0) = 1$.

参考教材例题 1，第 398 页

Exercise 72

解题思路

用导数符号证明递增，再比较与端点处的值。

SOLUTION

For (a), $f'(x) = 1 - 1/x > 0$ for $x > 1$.

For (b), $f(x) > f(1) = 1$, hence $\ln x < x - 1 < x$.

参考教材6.1 节例题 7，第 336 页；例题 3，第 401 页

Exercise 77

解题思路

绕纵轴用水平圆盘，平方半径后用对数积分。

SOLUTION

The radius at y is $2/\sqrt{y+1}$. Therefore

$$V = \pi \int_0^3 \frac{4}{y+1}\,dy = 4\pi \ln 4.$$

参考教材6.3 节例题 2，第 352 页

Exercise 81

解题思路

两道题的导数都是两个正项之差，平方根可用它们的和化简。

SOLUTION

For (a), $y' = x/4 - 1/x$ and $\sqrt{1 + (y')^2} = x/4 + 1/x$. Hence $L = [x^2/8 + \ln x]_4^8 = 6 + \ln 2$.

For (b), $dx/dy = y/8 - 2/y$, so the length is $[y^2/16 + 2\ln y]_4^{12} = 8 + 2\ln 3$.

参考教材线性化公式，第 192 页；例题 1，第 398 页

Exercise 89

解题思路

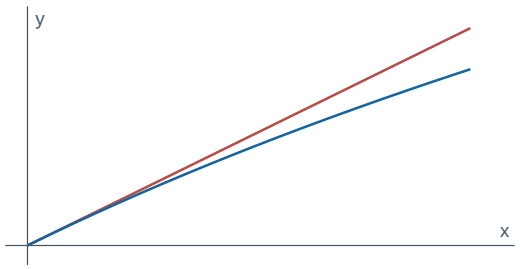
误差用直线减曲线，求导就能知道误差在哪里最大。

SOLUTION

For (a), $f(0) = 0$ and $f'(0) = 1$, so the linearization is $L(x) = x$.

(b) The error $E = x - \ln(1+x)$ has derivative $x/(1+x) \geq 0$ on $[0, 0.1]$. The maximum is $0.1 - \ln 1.1 \approx 0.00469$.

(c) The graphs are closest at 0 and farthest apart at 0.5. The maximum gap is $0.5 - \ln 1.5 \approx 0.0945349$.



Blue: ln(1 + x) Red: linearization

7.3 Exponential Functions

28 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 412–414

参考教材例题 1，第 405 页

Exercise 1

解题思路

两边取自然对数，解指数中的线性方程。

SOLUTION

Taking logarithms yields (a) $t = -\ln 27/0.3$, (b) $t = -\ln 2/k$ for $k \neq 0$, and (c) $t = \ln 0.4/\ln 0.2$. For (b), $k = 0$ gives no solution.

参考教材例题 3，第 406 页

Exercise 8

解题思路

对指数内的线性函数用链式法则。

SOLUTION

For $y = e^{2x/3}$, $y' = \frac{2}{3}e^{2x/3}$.

参考教材例题 3，第 406 页

Exercise 16

解题思路

先利用对数性质消去指数。

SOLUTION

On $u > 0$, $y = \ln 3 + \ln u - u$, hence $dy/du = 1/u - 1$.

参考教材例题 3，第 406 页

Exercise 18

解题思路

用乘积法则分别求导，再提公因子。

SOLUTION

For $y = \theta^3 e^{-2\theta}\cos 5\theta$,

$$y' = e^{-2\theta}\theta^2[(3-2\theta)\cos 5\theta - 5\theta \sin 5\theta].$$

参考教材5.4 节例题 2，第 298 页

Exercise 26

解题思路

两个变动端点分别应用基本定理，下限的贡献带负号。

SOLUTION

For $x > 0$, the upper contribution is $\ln(e^{2x})\,2e^{2x} = 4xe^{2x}$. The lower contribution is $\ln(e^{4\sqrt{x}})\,2e^{4\sqrt{x}}/\sqrt{x} = 8e^{4\sqrt{x}}$. Hence $y' = 4xe^{2x} - 8e^{4\sqrt{x}}$. At 0 its right derivative is -8.

参考教材例题 3，第 406 页；隐函数求导，第 159 页

Exercise 28

解题思路

两边隐式求导，把含导数的项移到一侧。

SOLUTION

From $\ln(xy) = e^{x+y}$,

$$\frac{1}{x} + \frac{y'}{y} = e^{x+y}(1 + y').$$

Thus $y' = (e^{x+y} - 1/x)/(1/y - e^{x+y})$ wherever the denominator is nonzero.

参考教材例题 3，第 406 页；隐函数求导，第 159 页

Exercise 29

解题思路

用链式法则后解出斜率。

SOLUTION

Differentiation gives $2e^{2x} = \cos(x+3y)(1+3y')$. Thus $y' = \frac{1}{3}[2e^{2x}/\cos(x+3y) - 1]$ where $\cos(x+3y) \neq 0$.

参考教材例题 4，第 407 页

Exercise 33

解题思路

对每个指数项积分，除以内层系数。

SOLUTION

The integral is $e^{3x}/3 - 5e^{-x} + C$, since differentiation recovers the two terms.

参考教材例题 4，第 407 页

Exercise 42

解题思路

取根号为新变量，分母中的根号随微分消去。

SOLUTION

Let $u = \sqrt{r}$, so $dr/\sqrt{r} = 2\,du$. The integral is $2\int e^{-u}\,du = -2e^{-\sqrt{r}} + C$.

参考教材例题 4，第 407 页

Exercise 52

解题思路

整体换成指数函数，并将端点映射到 1 与圆周率。

SOLUTION

Set $u = e^{x^2}$, so $du = 2xe^{x^2}\,dx$. The integral becomes $\int_1^{\pi} \cos u\,du = -\sin 1$.

参考教材例题 6，第 411 页

Exercise 60

解题思路

常数底的指数求导要乘以底数的自然对数。

SOLUTION

Writing $3^{-x} = e^{-x\ln 3}$ gives $y' = -(\ln 3)3^{-x}$.

参考教材例题 5，第 409 页

Exercise 63

解题思路

把实数幂写成自然指数的复合。

SOLUTION

For $x > 0$, $x^{\pi} = e^{\pi \ln x}$, so $y' = \pi x^{\pi-1}$.

参考教材例题 5，第 409 页

Exercise 66

解题思路

对外层实数幂和内层对数连续应用链式法则。

SOLUTION

On $u > 1$,

$$\frac{dy}{du} = \frac{\pi(\ln u)^{\pi-1}}{u}.$$

参考教材例题 6，第 411 页

Exercise 69

解题思路

先对指数函数求导，再乘正弦内层的导数。

SOLUTION

For $y = 2^{\sin 3t}$, $dy/dt = 3(\ln 2)2^{\sin 3t}\cos 3t$.

参考教材例题 7，第 412 页

Exercise 78

解题思路

先把平方根改写为二分之一次幂，再换底求导。

SOLUTION

On the real domain where $7x/(3x+2) > 0$,

$$y = \frac{1}{2}\ln\frac{7x}{3x+2}.$$

Thus $y' = \frac{1}{2}(1/x - 3/(3x+2)) = 1/[x(3x+2)]$.

参考教材例题 7，第 412 页

Exercise 79

解题思路

用乘积法则和以 7 为底的对数求导公式。

SOLUTION

For $u > 0$,

$$y' = \sin(\log_7 u) + \frac{\cos(\log_7 u)}{\ln 7}.$$

参考教材例题 7，第 412 页

Exercise 86

解题思路

利用换底把嵌套的指数与对数直接化简。

SOLUTION

Since $e^{(\sin t)\ln 3} = 3^{\sin t}$, $y = t\sin t$. Thus $y' = \sin t + t\cos t$.

参考教材例题 6，第 411 页

Exercise 89

解题思路

积分常数底指数时，除以指数的线性系数及底数的对数。

SOLUTION

An antiderivative of $2^{-\theta}$ is $-2^{-\theta}/\ln 2$. The integral from 0 to 1 is $1/(2\ln 2)$.

参考教材例题 6，第 411 页

Exercise 92

解题思路

取根号为新变量，微分恰好吸收分母。

SOLUTION

Let $u = \sqrt{x}$. Then $dx/\sqrt{x} = 2\,du$, and

$$2\int_1^2 2^u\,du = \frac{4}{\ln 2}.$$

参考教材例题 5，第 409 页

Exercise 95

解题思路

用对数求导倒推原函数，注意指数前有一个 2。

SOLUTION

For $x > 0$, $(x^{2x})' = 2(1+\ln x)x^{2x}$. The integral is $\frac{1}{2}(4^8 - 2^4) = 32760$.

参考教材例题 6，第 411 页

Exercise 100

解题思路

幂指数加一为自然对数 2，代入上下限即可。

SOLUTION

An antiderivative is $x^{\ln 2}/\ln 2$. The integral is $(e^{\ln 2} - 1)/\ln 2 = 1/\ln 2$.

参考教材例题 7，第 412 页

Exercise 103

解题思路

先换底，再识别自然对数平方的导数。

SOLUTION

Since $(\ln 2)\log_2 x = \ln x$, the integral is

$$\left[\tfrac{1}{2}(\ln x)^2\right]_1^4 = 2(\ln 2)^2.$$

参考教材例题 5，第 409 页

Exercise 115

解题思路

底数与指数都含变量，先取自然对数。

SOLUTION

On $x > -1$, $\ln y = x\ln(x+1)$. Thus

$$y' = (x+1)^x[\ln(x+1) + x/(x+1)].$$

参考教材例题 5，第 409 页

Exercise 122

解题思路

把双变量幂取对数，展开乘积导数。

SOLUTION

For $x > 1$, $\ln y = (\ln x)\ln(\ln x)$. Therefore

$$y' = \frac{(\ln x)^{\ln x}}{x}[\ln(\ln x) + 1].$$

参考教材导数判别法；例题 3，第 406 页

Exercise 131

解题思路

用一阶导数定极值、二阶导数找凹凸变化，并检查远处的趋势。

SOLUTION

Here $f' = e^{-x}(1 - x)$ and $f'' = e^{-x}(x - 2)$.

(a) The derivative changes from positive to negative at 1, giving absolute maximum $1/e$. As $x \to -\infty$, $f(x) \to -\infty$, so there is no absolute minimum.

(b) The concavity changes at 2, so the sole inflection point is $(2, 2e^{-2})$.

参考教材对数求导法；例题 3，第 406 页

Exercise 133

解题思路

改写为负的平方乘对数，再看导数在临界点两侧的符号。

SOLUTION

On $x > 0$, $f = -x^2 \ln x$, so $f' = -x(2\ln x + 1)$. This is positive below $e^{-1/2}$ and negative above it. The absolute maximum is $1/(2e)$ at $x = e^{-1/2}$.

参考教材6.3 节例题 1，第 352 页

Exercise 141

解题思路

弧长根式化为正割，再用其标准原函数。

SOLUTION

Since $y' = -\tan x$ and $\cos x > 0$, the length is

$$\int_0^{\pi/4} \sec x\, dx = \ln(\sqrt{2} + 1).$$

参考教材例题 3，第 406 页；图见第 414 页

Exercise 146

解题思路

用凸函数的中点切线作下界，用连接两端点的割线作上界。

SOLUTION

For (a), $(e^x)'' = e^x > 0$.

(b) Put $A = \ln a$, $B = \ln b$, $m = (A + B)/2$. Strict convexity puts the curve strictly above its tangent at m except at m, and below the endpoint chord in (A, B). Integrating the tangent gives $(B-A)e^m$, since its linear term about m integrates to zero. Integrating the chord gives $(B - A)(e^A + e^B)/2$.

(c) The middle integral is $b-a$. Divide by $B-A > 0$ and use $e^m = \sqrt{ab}$ to obtain the desired strict inequalities.

7.5 Indeterminate Forms and L'HÔpital's Rule

20 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 432–433

参考教材例题 1，第 426 页

Exercise 16

解题思路

每次使用洛必达法前都检查是否仍为零比零。

SOLUTION

The successive derivative quotients are $(\cos x - 1)/(3x^2)$, $-\sin x/(6x)$, and $-\cos x/6$. Each preceding quotient is of type $0/0$, so the limit is $-1/6$.

参考教材例题 1，第 426 页

Exercise 20

解题思路

分子分母在 1 处同时为零，求导后分母非零。

SOLUTION

The derivative quotient is $1/(1/x - \pi\cos \pi x)$, giving $1/(1+\pi)$.

参考教材例题 1，第 426 页

Exercise 21

解题思路

对数的导数化为正切，再求一次导。

SOLUTION

L'Hôpital's Rule first gives $\lim_{x\to 0} 2x/\tan x$, still of type $0/0$. Applying it again gives $2/\sec^2(0) = 2$.

参考教材例题 5，第 428 页

Exercise 26

解题思路

令与端点的距离为新变量，把乘积改写成标准正弦比值。

SOLUTION

Put $h = \pi/2 - x \to 0^+$. Then $h\tan x = (h/\sin h)\cos h \to 1$. The quotient $h/\sin h$ also gives 1 by l'Hôpital's Rule.

参考教材例题 4，第 428 页

Exercise 31

解题思路

先换底，再处理无穷比无穷。

SOLUTION

By change of base the ratio is $(\ln 2)\ln(x+1)/\ln x$. The derivative ratio of the logarithms is $x/(x+1) \to 1$. Thus the limit is $\ln 2$.

参考教材例题 6，第 428 页

Exercise 38

解题思路

合并两个对数，把无穷差化成真数的比值。

SOLUTION

For small positive x, the expression is $\ln(x/\sin x)$. The argument tends to 1 by l'Hôpital's Rule, so the limit is 0.

参考教材例题 6，第 428 页

Exercise 40

解题思路

先拆出常数，再通分处理两个倒数。

SOLUTION

Write the expression as $3 + (\sin x - x)/(x \sin x)$. The latter quotient is $0/0$. Its derivative quotient is $(\cos x - 1)/(\sin x + x \cos x)$, also $0/0$. Differentiating again gives a limit of 0. The answer is 3.

参考教材例题 6，第 428 页

Exercise 41

解题思路

通分后仍是零比零，需要连续求导两次。

SOLUTION

The quotient is $(\ln x - x + 1)/[(x - 1) \ln x]$. Both numerator and denominator, and then their first derivatives, vanish at 1. The second derivatives are $-1/x^2$ and $1/x + 1/x^2$, giving $-1/2$.

参考教材例题 4，第 428 页

Exercise 45

解题思路

指数项增长快于一次和二次项，可通过两次求导消去幂次。

SOLUTION

The first derivative ratio is $(e^t + 2t)/(e^t - 1)$, still of type ∞/∞. The second is $(e^t + 2)/e^t \to 1$.

参考教材例题 7，第 429 页

Exercise 51

解题思路

先求幂表达式的对数极限，再取指数。

SOLUTION

The logarithm is $\ln x/(1 - x)$, of type $0/0$. Its limit is -1 by the derivative quotient $-1/x$. Thus the original limit is e^{-1}.

参考教材例题 8，第 430 页

Exercise 53

解题思路

把幂化为自然指数，比较二重对数与横坐标的增长速度。

SOLUTION

The logarithm is $\ln(\ln x)/x$. The derivative quotient is $1/(x \ln x) \to 0$, so the limit is 1.

参考教材例题 7，第 429 页

Exercise 59

解题思路

指数的对数为乘积，改写成一个商。

SOLUTION

The logarithm is $x \ln x = \ln x/(1/x)$. L'Hôpital's Rule gives the limit of $-x$, namely 0. Therefore $x^x \to 1$.

参考教材例题 7，第 429 页

Exercise 60

解题思路

先对整个幂取对数，用倒数把乘积改成商。

SOLUTION

The logarithm is $\ln(1 + 1/x)/(1/x)$ as $x \to 0^+$. The derivative quotient equals $x/(x + 1) \to 0$. Thus the limit is 1.

参考教材例题 7，第 429 页

Exercise 61

解题思路

把底数写成一加小量，用对数的标准极限。

SOLUTION

Write $(x+2)/(x-1) = 1+h$, where $h = 3/(x-1)$. Then $x\ln(1+h) = (xh)[\ln(1+h)/h] \to 3$, since the logarithmic quotient tends to 1. The limit is e^3.

参考教材例题 5，第 428 页

Exercise 66

解题思路

把正弦拆成横坐标与有界比值，剩下已知的乘积极限。

SOLUTION

We have $\sin x \ln x = (\sin x/x)(x\ln x)$. The factors tend to 1 and 0, respectively, the latter by applying l'Hôpital's Rule to $\ln x/(1/x)$. The limit is 0.

参考教材根式的极限法则；例题 4，第 428 页

Exercise 67

解题思路

在根号内外提出最高次幂，避免循环求导。

SOLUTION

For $x > 0$, the ratio equals $\sqrt{9+1/x}/(1+1/\sqrt{x})$, which tends to 3.

参考教材标准三角极限；例题 5，第 428 页

Exercise 68

解题思路

先把比值合到一个根号里，再用正弦的基本极限。

SOLUTION

For $x > 0$ near zero, the ratio is $\sqrt{x/\sin x} \to 1$.

参考教材例题 2，第 427 页

Exercise 75

解题思路

直接代入分母即可判断能否使用洛必达法。

SOLUTION

Part (b) is correct: the denominator tends to 6 and the numerator to 0, so the quotient tends to 0. In (a), l'Hôpital's Rule does not apply because the original quotient is neither $0/0$ nor ∞/∞.

参考教材例题 7，第 429 页

Exercise 85

解题思路

对整式取对数，利用对数在零点附近的线性行为。

SOLUTION

For fixed r, the base is positive for large k. When $r \neq 0$,

$$k\ln(1+r/k) = r\frac{\ln(1+r/k)}{r/k} \to r.$$

Exponentiation yields e^r. If $r = 0$, each term is 1.

参考教材例题 4，第 428 页

Exercise 88

解题思路

从零点处导数定义出发，利用指数衰减压过任意幂。

SOLUTION

The difference quotient is $e^{-1/x^2}/x$. With $u = 1/x^2$, its magnitude is $\sqrt{u}/e^u \to 0$ as $u \to \infty$, by l'Hôpital's Rule. Hence $f'(0) = 0$.

7.8 Relative Rates of Growth

8 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 458–459

参考教材例题 1–2，第 455–456 页

Exercise 1

解题思路

各项除以自然指数函数，再看比值趋于零、常数还是无穷。

SOLUTION

The slower functions are (a), (b), (c), (e), (f), and (h): polynomial or logarithmic factors are dominated by e^x, and the ratios for (e) and (f) decay exponentially. The function in (g), $e^x/2$, grows at the same rate. The ratio in (d) is $(4/e)^x \to \infty$, so (d) grows faster.

参考教材例题 3–4，第 457 页

Exercise 9

解题思路

小 o 要求比值趋零，大 O 只要求比值最终有界。

SOLUTION

The ratios of the two comparison functions in (a)–(h) tend, respectively, to $1, 1, 1, 1/2, 0, 1, 1, \sqrt{2}$. Hence (a) F, (b) F, (c) T, (d) T, (e) T, (f) T, (g) F, (h) T.

参考教材例题 3–4，第 457 页

Exercise 10

解题思路

对各陈述先构造比值，再按有界和趋零分别判断。

SOLUTION

For (a) the ratio $1 + 3x$ is unbounded, so F. For (b), $1 + 1/x$ is bounded, so T. For (c), $1 - 1/x \to 1$, so F. For (d), $(2 + \cos x)/2$ is bounded, so T. For (e)–(h), the ratios tend to $1, 0, 0, 1/2$, respectively, giving T, T, T, F.

参考教材例题 2、4，第 456–457 页

Exercise 11

解题思路

同阶意味着比值趋向正有限常数，倒数也如此。

SOLUTION

If $f/g \to c$ with $0 < c < \infty$, then eventually $c/2 < f/g < 2c$. This bounds both f/g and g/f, so $f = O(g)$ and $g = O(f)$.

参考教材例题 3，第 457 页

Exercise 12

解题思路

非零多项式的最高次项决定比值的阶。

SOLUTION

If the nonzero polynomials have degrees m, n and leading coefficients a, b, then $(f/g)/x^{m-n} \to a/b \neq 0$. Thus $f = o(g)$ exactly when $m < n$. The zero polynomial is $o(g)$ for every nonzero g.

参考教材例题 1，第 455 页

Exercise 20

解题思路

多项式逐项与指数比较，每次求导降低一个幂次。

SOLUTION

For each integer $j \geq 0$, $x^j/e^x \to 0$, by repeated l'Hôpital for $j > 0$. Hence $p(x)/e^x = \sum_{j=0}^{n} a_j x^j/e^x \to 0$ for every polynomial p.

参考教材例题 1，第 455 页

Exercise 21

解题思路

只做 a。对无穷比无穷使用一次洛必达。

SOLUTION

For every positive integer n,

$$\lim_{x\to\infty}\frac{\ln x}{x^{1/n}}=\lim_{x\to\infty}\frac{n}{x^{1/n}}=0.$$

Thus $\ln x=o(x^{1/n})$, including $n=1000000$.

参考教材例题 1，第 455 页

Exercise 22

解题思路

非恒定多项式最终与它的最高次幂成固定比例。

SOLUTION

Let p have degree $n\geq 1$ and leading coefficient $a_n\neq 0$. Since $\ln x/x^n\to 0$ by l'Hôpital and $p(x)/x^n\to a_n$, the quotient $\ln x/p(x)$ tends to zero.

7.6 Inverse Trigonometric Functions

24 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 442–445

参考教材例题 3，第 438 页

Exercise 1

解题思路

在反正切的主值范围内寻找标准角。

SOLUTION

In $(-\pi/2, \pi/2)$ the required angles are (a) $\pi/4$, (b) $-\pi/3$, and (c) $\pi/6$.

参考教材例题 1，第 435 页

Exercise 3

解题思路

反正弦的主值落在正负半个圆周率之间。

SOLUTION

The principal angles with the given sines are (a) $-\pi/6$, (b) $\pi/4$, and (c) $-\pi/3$.

参考教材例题 1，第 435 页

Exercise 5

解题思路

用反余弦的主值区间筛选标准角。

SOLUTION

In $[0, \pi]$ the required angles are (a) $\pi/3$, (b) $3\pi/4$, and (c) $\pi/6$.

参考教材例题 1，第 435 页；图 7.31，第 440 页

Exercise 7

解题思路

先取倒数，再在反余弦的主值区间取角。

SOLUTION

Using $\operatorname{arcsec} x = \arccos(1/x)$ gives (a) $3\pi/4$, (b) $\pi/6$, and (c) $2\pi/3$.

参考教材例题 1，第 435 页

Exercise 13

解题思路

反正弦在定义域的右端点左连续。

SOLUTION

By continuity on $[-1, 1]$, the limit is $\arcsin 1 = \pi/2$.

参考教材例题 3，第 438 页

Exercise 16

解题思路

反正切的左端极限由其水平渐近线决定。

SOLUTION

As $x \to -\infty$, the principal angle with tangent x tends to $-\pi/2$ from above.

参考教材图 7.31，第 440 页

Exercise 18

解题思路

把反正割改写为反余弦，倒数趋于零。

SOLUTION

Since $\operatorname{arcsec} x = \arccos(1/x)$, continuity gives the limit $\arccos 0 = \pi/2$.

参考教材例题 4，第 439 页

Exercise 22

解题思路

内层倒数的负号与反余弦导数的负号抵消。

SOLUTION

For $|x| > 1$, the chain rule gives

$$y' = \frac{1}{x^2\sqrt{1 - 1/x^2}} = \frac{1}{|x|\sqrt{x^2 - 1}}.$$

参考教材例题 5，第 440 页

Exercise 25

解题思路

反正割求导要保留绝对值，并乘以内层导数。

SOLUTION

For $|2s + 1| > 1$,

$$\frac{dy}{ds} = \frac{2}{|2s + 1|\sqrt{(2s + 1)^2 - 1}}.$$

参考教材例题 4，第 439 页

Exercise 33

解题思路

对外层对数和内层反正切分别求导。

SOLUTION

The logarithm requires $x > 0$. Its derivative is $y' = 1/[(1 + x^2)\arctan x]$.

参考教材例题 6，第 441 页

Exercise 48

解题思路

根式化到反正弦的标准形式。

SOLUTION

With $u = 2x$, the integral is $\frac{1}{2}\int du/\sqrt{1 - u^2} = \frac{1}{2}\arcsin(2x) + C$.

参考教材例题 6，第 441 页

Exercise 51

解题思路

提出根号内的常数，使剩余积分符合反正割的形式。

SOLUTION

Set $u = 5x/\sqrt{2}$. The integral becomes $\frac{1}{\sqrt{2}}\int du/(u\sqrt{u^2 - 1})$. Thus it equals $\frac{1}{\sqrt{2}}\operatorname{arcsec}|5x/\sqrt{2}| + C$ on each interval of definition.

参考教材例题 6，第 441 页

Exercise 56

解题思路

缩放变量，得到反正切标准积分，再利用对称性。

SOLUTION

An antiderivative is $\frac{1}{2\sqrt{3}}\arctan(\sqrt{3}\,t/2)$. The endpoint angles are $\pm\pi/3$, giving $\pi/(3\sqrt{3})$.

参考教材例题 6，第 441 页

Exercise 67

解题思路

指数函数整体换元，上下限同时变换。

SOLUTION

Set $u = e^x$, $du = e^x\,dx$. The integral is $\arctan\sqrt{3} - \arctan 1 = \pi/12$.

参考教材例题 6，第 441 页

Exercise 69

解题思路

根式中的四次幂可视作平方的平方。

SOLUTION

Let $u = y^2$, so $du = 2y\,dy$. The integral becomes $\frac{1}{2}\int du/\sqrt{1 - u^2} = \frac{1}{2}\arcsin(y^2) + C$.

参考教材例题 7，第 442 页

Exercise 73

解题思路

先配方，把根式变成半圆标准形式。

SOLUTION

Since $3 - 2t - t^2 = 4 - (t+1)^2$, an antiderivative is $6\arcsin((t+1)/2)$. Its increment from -1 to 0 is $6(\pi/6) = \pi$.

参考教材例题 7，第 442 页

Exercise 78

解题思路

配方后识别以 3 为中心的反正切积分。

SOLUTION

The denominator is $(x-3)^2 + 1$. The integral is $2[\arctan(1) - \arctan(-1)] = \pi$.

参考教材例题 6，第 441 页

Exercise 91

解题思路

利用正割与反正割的复合关系，约去多余因子。

SOLUTION

For $x \in [\sqrt{2}, 2]$, $\sec^2(\operatorname{arcsec} x) = x^2$. The integrand reduces to $x/\sqrt{x^2-1}$. Thus the integral is $\left[\sqrt{x^2-1}\right]_{\sqrt{2}}^{2} = \sqrt{3} - 1$.

参考教材7.5 节例题 1，第 426 页；例题 4，第 439 页

Exercise 97

解题思路

检查零比零，再对分子使用反正弦求导公式。

SOLUTION

The derivative quotient is $5/\sqrt{1-25x^2} \to 5$. The limit is 5.

参考教材7.5 节例题 1，第 426 页；例题 5，第 440 页

Exercise 98

解题思路

对正侧的反正割与根式分别求导，根式因子抵消。

SOLUTION

For $x > 1$, the derivative quotient is

$$\frac{x/\sqrt{x^2-1}}{1/(x\sqrt{x^2-1})} = x^2.$$

Both original terms tend to zero, so l'Hôpital's Rule gives 1.

参考教材例题 3，第 438 页

Exercise 102

解题思路

反正切有界，先把分子分母中的指数约到适合比较的形式。

SOLUTION

For $x > 0$, the ratio is nonnegative and at most $(\pi/2)e^{-x}$, since the denominator is at least e^{2x}. The limit is 0.

参考教材7.5 节例题 1，第 426 页；例题 4，第 439 页

Exercise 103

解题思路

令根号为新变量，把平方拆成标准反正切比值的平方。

SOLUTION

For $u = \sqrt{x} \to 0^+$ the expression is $(\arctan u/u)^2/\sqrt{1+u^2}$. Since the derivative of arctan at zero is 1, the limit is 1.

参考教材6.1 节例题 1，第 333 页；例题 6，第 441 页

Exercise 129

解题思路

同一纵向跨度分别是圆的直径和正方形的边长。

SOLUTION

The span is $d(x) = 2/\sqrt{1+x^2}$. For (a), the area is $\pi d^2/4$, so $V = \pi[\arctan x]_{-1}^{1} = \pi^2/2$. For (b), the area is d^2, so $V = 4[\arctan x]_{-1}^{1} = 2\pi$.

参考教材例题 3，第 438 页

Exercise 133

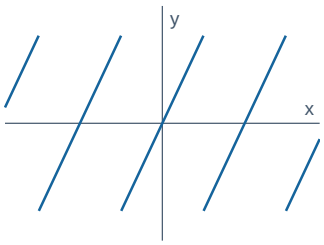
解题思路

反正切只能返回主值，因此第一种复合会折回到主值区间。

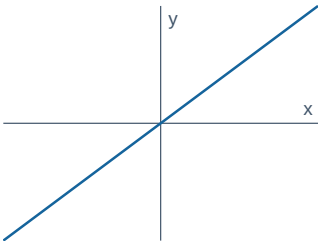
SOLUTION

For (a), exclude $x = \pi/2 + k\pi$ for $k \in \mathbb{Z}$. On $(-\pi/2 + k\pi, \pi/2 + k\pi)$, the graph is the segment $y = x - k\pi$. Its range is $(-\pi/2, \pi/2)$.

For (b), the domain and range are both $\mathbb{R}$, and the graph is $y = x$. Here $\arctan x$ always lies in the branch where tangent is its inverse.



(a) arctan(tan x)



(b) tan(arctan x)

8.2 Integration by Parts

18 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 475–477

参考教材例题 1，第 471 页

Exercise 2

解题思路

让一次因子求导，让余弦积分。

SOLUTION

Take $U = u$ and $dV = \cos(\pi u)\,du$. Integration by parts gives

$$\frac{u\sin(\pi u)}{\pi} + \frac{\cos(\pi u)}{\pi^2} + C.$$

参考教材例题 2，第 472 页

Exercise 6

解题思路

让对数求导，幂函数积分，再代入两端。

SOLUTION

With $U = \ln x$ and $dV = x^3\,dx$, the antiderivative is $x^4\ln x/4 - x^4/16$. Its increment from 1 to e is $(3e^4 + 1)/16$.

参考教材例题 2，第 472 页

Exercise 12

解题思路

把反正弦作为求导的因子，另一因子为 1。

SOLUTION

Integration by parts gives

$$y\arcsin y - \int \frac{y\,dy}{\sqrt{1-y^2}} = y\arcsin y + \sqrt{1-y^2} + C.$$

参考教材例题 3，第 473 页

Exercise 17

解题思路

多项式逐次求导降次，指数函数保持原形。

SOLUTION

For $P = x^2 - 5x$, repeated integration by parts gives $e^x(P - P' + P'') + C = e^x(x^2 - 7x + 7) + C$.

参考教材例题 4，第 473 页

Exercise 23

解题思路

分部两次后出现原积分，移项解出它。

SOLUTION

Write $I = \int e^{2x}\cos 3x\,dx$ and $J = \int e^{2x}\sin 3x\,dx$. Parts gives $I = e^{2x}\cos 3x/2 + 3J/2$ and $J = e^{2x}\sin 3x/2 - 3I/2$, up to constants. Solving gives

$$I = \frac{e^{2x}}{13}(2\cos 3x + 3\sin 3x) + C.$$

参考教材例题 2，第 472 页

Exercise 28

解题思路

先拆对数，并在第二项平移变量，再分别分部积分。

SOLUTION

On $x < -1$ or $x > 0$, write $\ln(x + x^2) = \ln|x| + \ln|x + 1|$. In the second term set $u = x + 1$. Integrating both logarithms by parts gives

$$x\ln|x| + (x+1)\ln|x+1| - 2x + C.$$

参考教材例题 4，第 473 页

Exercise 29

解题思路

用对数作新变量，得到指数乘正弦，再分部两次。

SOLUTION

Let $u = \ln x$, so $dx = e^u\,du$. Twice integrating $\int e^u \sin u\,du$ by parts gives $e^u(\sin u - \cos u)/2 + C$. Hence the answer is $x[\sin(\ln x) - \cos(\ln x)]/2 + C$.

参考教材换元法；5.5 节例题 1，第 308 页

Exercise 31

解题思路

有内层平方及其导数，先做换元。

SOLUTION

With $u = x^2$, the integral is $\frac{1}{2}\int \sec u\,du = \frac{1}{2}\ln|\sec(x^2) + \tan(x^2)| + C$.

参考教材换元法；5.5 节例题 4，第 310 页

Exercise 32

解题思路

取平方根作新变量，分母随微分消去。

SOLUTION

Set $u = \sqrt{x}$, so $dx/\sqrt{x} = 2\,du$. Then the result is $2\int \cos u\,du = 2\sin\sqrt{x} + C$.

参考教材换元法；7.3 节例题 4，第 407 页

Exercise 37

解题思路

四次幂的微分与外部三次项匹配。

SOLUTION

With $u = x^4$, the integral becomes $\frac{1}{4}\int e^u\,du = e^{x^4}/4 + C$.

参考教材例题 2，第 472 页

Exercise 43

解题思路

取对数为求导因子，剩下的根号幂积分。

SOLUTION

Take $U = \ln x$ and $dV = \sqrt{x}\,dx$. Then

$$\int \sqrt{x}\ln x\,dx = \frac{2}{3}x^{3/2}\ln x - \frac{4}{9}x^{3/2} + C.$$

参考教材例题 1，第 471 页

Exercise 45

解题思路

取根号换元后，变成线性因子乘余弦。

SOLUTION

Set $u = \sqrt{x}$, so $dx = 2u\,du$. Parts gives $2\int u\cos u\,du = 2u\sin u + 2\cos u + C$. Substitute $u = \sqrt{x}$.

参考教材例题 2，第 472 页

Exercise 50

解题思路

先令平方为新变量，再对反正弦做分部积分。

SOLUTION

Set $u = x^2$. The bounds become $0, 1/2$. An antiderivative of $\arcsin u$ is $u\arcsin u + \sqrt{1-u^2}$ by parts. Its increment is $\pi/12 + \sqrt{3}/2 - 1$.

参考教材例题 2，第 472 页

Exercise 51

解题思路

反正切求导后可用多项式除法化简分式。

SOLUTION

With $U = \arctan x$ and $dV = x\,dx$, the remainder is $\frac{1}{2}\int x^2/(1+x^2)\,dx$. Using $x^2/(1+x^2) = 1 - 1/(1+x^2)$ gives $[(x^2+1)\arctan x - x]/2 + C$.

参考教材乘积法则；7.3 节例题 3，第 406 页

Exercise 53

解题思路

识别原式恰为一个乘积的导数。

SOLUTION

Since $(xe^{x^2})' = e^{x^2} + 2x^2e^{x^2}$, the integral equals $xe^{x^2} + C$.

参考教材例题 4，第 473 页

Exercise 65

解题思路

对指数乘余弦积分，最后除以区间长度。

SOLUTION

Integration by parts twice gives the antiderivative $e^{-t}(\sin t - \cos t)$ for $2e^{-t}\cos t$. Therefore the average is $(1 - e^{-2\pi})/(2\pi)$.

参考教材例题 5，第 474 页

Exercise 67

解题思路

把幂函数降一次，直接得到递推关系。

SOLUTION

Choose $U = x^n$, $dV = \cos x\,dx$. Then $dU = nx^{n-1}dx$ and $V = \sin x$. Integration by parts yields

$$\int x^n \cos x\,dx = x^n \sin x - n\int x^{n-1}\sin x\,dx.$$

参考教材例题 2，第 472 页；5.4 节例题 2，第 298 页

Exercise 75

解题思路

把内层定积分当作求导因子，选择原函数让两端边界项都为零。

SOLUTION

Let $F(x) = \int_x^b f(t)\,dt$, so $F' = -f$ for continuous f. Integrate F by parts with $V = x - a$. The endpoint term $[(x-a)F(x)]_a^b$ vanishes, and the remaining integral is $\int_a^b (x-a)f(x)\,dx$.

8.3 Trigonometric Integrals

16 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 483–484

参考教材例题 1，第 479 页

Exercise 3

解题思路

把单独的正弦因子作为余弦的微分。

SOLUTION

With $u = \cos x$, the integral is $-\int u^3\,du = -\cos^4 x/4 + C$.

参考教材例题 3，第 480 页

Exercise 14

解题思路

偶次正弦用半角公式降次。

SOLUTION

Since $\sin^2 x = (1 - \cos 2x)/2$, the integral is $[x/2 - \sin 2x/4]_0^{\pi/2} = \pi/4$.

参考教材例题 1，第 479 页

Exercise 15

解题思路

留一个正弦因子作微分，把其余偶次幂改成余弦。

SOLUTION

Set $u = \cos y$. Then the integral is $\int_0^1 (1 - u^2)^3\,du$. Expanding gives $1 - 1 + 3/5 - 1/7 = 16/35$.

参考教材例题 3，第 480 页

Exercise 19

解题思路

先用二倍角合并乘积，再用半角降次。

SOLUTION

The integrand is $4\sin^2(2x) = 2 - 2\cos 4x$. An antiderivative is $2x - \sin 4x/2 + C$.

参考教材例题 3，第 480 页

Exercise 20

解题思路

把一对正弦平方与余弦平方合并，再把剩余正弦平方降次。

SOLUTION

The integrand is $2\sin^2 y \sin^2(2y)$, which equals $(1 - \cos 2y)(1 - \cos 4y)/2$. All cosine terms, including $\cos 2y \cos 4y = (\cos 2y + \cos 6y)/2$, integrate to zero on $[0, \pi]$. The value is $\pi/2$.

参考教材例题 4，第 480 页

Exercise 24

解题思路

化成正弦平方再开方，注意在给定区间上的符号。

SOLUTION

Since $1 - \cos 2x = 2\sin^2 x$ and $\sin x \geq 0$ on $[0, \pi]$, the integral is $\sqrt{2}\int_0^\pi \sin x\,dx = 2\sqrt{2}$.

参考教材例题 4，第 480 页

Exercise 27

解题思路

半角公式可消去根式，再用半角余弦换元。

SOLUTION

On the stated interval the integrand becomes $2\sqrt{2}\sin(x/2)\cos^2(x/2)$. Therefore

$$\left[-\frac{4\sqrt{2}}{3}\cos^3(x/2)\right]_{\pi/3}^{\pi/2} = \frac{\sqrt{6}}{2} - \frac{2}{3}.$$

参考教材例题 4，第 480 页

Exercise 30

解题思路

根式是正弦减余弦的绝对值，用区间判断正负。

SOLUTION

On $[\pi/2, 3\pi/4]$, $\sqrt{1-\sin 2x} = \sin x - \cos x$. Thus the integral is $[-\cos x - \sin x]_{\pi/2}^{3\pi/4} = 1$.

参考教材例题 7，第 482 页

Exercise 33

解题思路

把正割平方作为正切的微分。

SOLUTION

With $u = \tan x$, the integral is $\int u\,du = \tan^2 x/2 + C$.

参考教材例题 6，第 481 页

Exercise 34

解题思路

把正切平方换成正割平方减一，套用正割立方公式。

SOLUTION

The integrand equals $\sec^3 x - \sec x$. Using the parts formula for $\int \sec^3 x\,dx$ gives

$$\tfrac{1}{2}[\sec x \tan x - \ln|\sec x + \tan x|] + C.$$

参考教材例题 7，第 482 页

Exercise 35

解题思路

留出正割乘正切，作为正割的微分。

SOLUTION

With $u = \sec x$, the integral becomes $\int u^2\,du = \sec^3 x/3 + C$.

参考教材例题 8，第 483 页

Exercise 53

解题思路

两频率相同，积化和差后留下常数项。

SOLUTION

The integrand is $\sin^2(3x) = (1-\cos 6x)/2$. On $[-\pi, \pi]$ the cosine integrates to zero, so the value is π.

参考教材例题 8，第 483 页

Exercise 54

解题思路

正弦与余弦的乘积化成二倍角。

SOLUTION

Since $\sin x \cos x = \frac{1}{2}\sin 2x$, the integral is $[-\cos 2x/4]_0^{\pi/2} = 1/2$.

参考教材例题 8，第 483 页

Exercise 56

解题思路

不同频率的余弦乘积化为两个余弦。

SOLUTION

The product equals $(\cos 6x + \cos 8x)/2$. Each term integrates to zero on $[-\pi/2, \pi/2]$, so the integral is 0.

参考教材例题 1，第 479 页

Exercise 64

解题思路

留出一个正弦，将正弦平方换成余弦平方。

SOLUTION

Let $u = \cos x$. The integral becomes $-\int (1 - u^2)u^{-4}\,du = 1/(3u^3) - 1/u + C$. Hence the answer is $\sec^3 x/3 - \sec x + C$.

参考教材例题 3，第 480 页；8.2 节例题 1，第 471 页

Exercise 67

解题思路

先对正弦平方降次，再对带有一次因子的余弦项分部积分。

SOLUTION

Using $\sin^2 x = (1 - \cos 2x)/2$, the integral is $x^2/4 - \frac{1}{2}\int x\cos 2x\,dx$. Integration by parts gives

$$\frac{x^2}{4} - \frac{x\sin 2x}{4} - \frac{\cos 2x}{8} + C.$$

8.4 Trigonometric Substitutions

14 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 488–489

参考教材例题 1，第 485 页

Exercise 2

解题思路

用正切替换根式内的线性项，把平方和化为正割平方。

SOLUTION

Set $3x = \tan\theta$ with $|\theta| < \pi/2$. The integral becomes $\int \sec\theta\, d\theta$. Thus the result is $\ln(3x + \sqrt{1+9x^2}) + C$.

参考教材例题 1，第 485 页

Exercise 3

解题思路

用正切换元后，分母和微分中的正割平方抵消。

SOLUTION

With $x = 2\tan\theta$, the integrand becomes $\frac{1}{2}d\theta$ and the bounds become $-\pi/4, \pi/4$. The value is $\pi/4$.

参考教材例题 3，第 486 页

Exercise 5

解题思路

用正弦替代横坐标，直接换算角度上下限。

SOLUTION

Set $x = 3\sin\theta$, so $dx/\sqrt{9-x^2} = d\theta$ on the given interval. The angle ranges from 0 to $\pi/6$, giving $\pi/6$.

参考教材例题 3，第 486 页

Exercise 8

解题思路

正弦换元将根式变成余弦，再用半角公式。

SOLUTION

Set $3t = \sin\theta$, giving $\frac{1}{3}\int \cos^2\theta\, d\theta$. Integrating and back-substituting yields

$$\frac{t}{2}\sqrt{1-9t^2} + \frac{1}{6}\arcsin(3t) + C.$$

参考教材例题 4，第 487 页

Exercise 9

解题思路

正割换元化简平方差根式。

SOLUTION

Put $2x = 7\sec\theta$ with $0 < \theta < \pi/2$. The integral becomes $\frac{1}{2}\int \sec\theta\, d\theta$. Back-substitution gives $\frac{1}{2}\ln(2x + \sqrt{4x^2-49}) + C$.

参考教材例题 4，第 487 页

Exercise 10

解题思路

把线性项换成正割，根式化为正切。

SOLUTION

With $5x = 3\sec\theta$, the integral becomes $\int \sec\theta\, d\theta$, hence $\ln(5x + \sqrt{25x^2-9}) + C$.

参考教材例题 3，第 486 页

Exercise 15

解题思路

正弦换元使根式与微分抵消，剩一个正弦。

SOLUTION

Set $x = 3\sin\theta$. Then the integral is $3\int \sin\theta\, d\theta = -3\cos\theta + C = -\sqrt{9-x^2} + C$.

参考教材例题 1，第 485 页

Exercise 17

解题思路

正切换元后，把正切平方改写成正割平方减一。

SOLUTION

Set $x = 2\tan\theta$. The integral is $8\int(\sec^2\theta - 1)\sec\theta\tan\theta\, d\theta$. With $u = \sec\theta$, this gives $8(u^3/3 - u) + C$, or $(x^2+4)^{3/2}/3 - 4\sqrt{x^2+4} + C$.

参考教材例题 3，第 486 页

Exercise 21

解题思路

先把根式化成含平方差的分式，再用正弦换元。

SOLUTION

For $-1 < x < 1$, the integrand is $(1+x)/\sqrt{1-x^2}$. With $x = \sin\theta$ it becomes $(1+\sin\theta)\, d\theta$. The result is $\arcsin x - \sqrt{1-x^2} + C$.

参考教材例题 4，第 487 页

Exercise 26

解题思路

正割换元后化为余割平方乘余割与余切的乘积。

SOLUTION

Set $x = \sec\theta$. The transformed integrand is $\cos\theta/\sin^4\theta$. Integrating gives $-1/(3\sin^3\theta) + C$, hence

$$-\frac{x^3}{3(x^2-1)^{3/2}} + C.$$

参考教材例题 1，第 485 页

Exercise 36

解题思路

先用指数换元，再用正切换元化简平方和。

SOLUTION

Let $u = e^t$, then $u = \tan\theta$. The transformed integral is $\int \cos\theta\, d\theta$, with primitive $u/\sqrt{1+u^2}$. The endpoint values are $4/5$ and $3/5$, so the integral is $1/5$.

参考教材7.6 节例题 6，第 441 页

Exercise 37

解题思路

从分母提出根号，并用两倍根号换元得到反正切。

SOLUTION

Let $u = 2\sqrt{t}$. The integral becomes $2\int_{1/\sqrt{3}}^{1} du/(1+u^2)$. Its value is $2(\pi/4 - \pi/6) = \pi/6$.

参考教材例题 3，第 486 页；8.2 节例题 1，第 471 页

Exercise 61

解题思路

三条路径都把原式化为容易积分的幂。

SOLUTION

Write $u = 1 - x^2$.

(a) Use parts with $U = x^2$, $dV = x\sqrt{1-x^2}\,dx$. The result is $-x^2u^{3/2}/3 - 2u^{5/2}/15 + C$.

(b) Directly substituting u gives $-\frac{1}{2}\int (1-u)u^{1/2}\,du = -u^{3/2}/3 + u^{5/2}/5 + C$.

(c) With $x = \sin\theta$, the integrand is $\sin^3\theta\cos^2\theta\,d\theta$. Let $v = \cos\theta$ to get $-v^3/3 + v^5/5 + C$. Since $v^2 = u$, all three forms agree.

参考教材例题 3，第 486 页

Exercise 62

解题思路

绳子是切线。先由直角三角形求斜率，再用正弦换元积分。

SOLUTION

For (a), the horizontal leg of the rope triangle is x and the upward leg is $\sqrt{100-x^2}$. From the skier to the boat the horizontal change is $-x$, so the slope is $-\sqrt{100-x^2}/x$.

For (b), set $x = 10\sin\theta$ to obtain $-10(\csc\theta - \sin\theta)\,d\theta$. Integrate and use $f(10) = 0$ to get

$$f(x) = 10\ln\frac{10+\sqrt{100-x^2}}{x} - \sqrt{100-x^2}$$

for $0 < x \le 10$.

8.5 Integration of Rational Functions by Partial Fractions

11 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 495–496

参考教材例题 1，第 490 页

Exercise 6

解题思路

先约去公因子，再拆成两个不同线性分母。

SOLUTION

On the original domain $z \neq 0, -2, 3$, cancel z. The decomposition is

$$\frac{1}{(z-3)(z+2)} = \frac{1}{5}\left(\frac{1}{z-3} - \frac{1}{z+2}\right).$$

参考教材例题 3，第 492 页

Exercise 7

解题思路

同次分式先做除法，余式再分解。

SOLUTION

Division gives $1 + (5t+2)/[(t-2)(t-3)]$. Clearing denominators and substituting $t = 2, 3$ yields

$$1 - \frac{12}{t-2} + \frac{17}{t-3}.$$

参考教材例题 1，第 490 页

Exercise 13

解题思路

用分母的两个零点求系数，再用对数原函数。

SOLUTION

The decomposition is $3/[4(y-3)] + 1/[4(y+1)]$. Evaluating the logarithms from 4 to 8 gives

$$\frac{3}{4}\ln 5 + \frac{1}{4}\ln(9/5) = \frac{1}{4}\ln 225.$$

参考教材例题 1，第 490 页

Exercise 16

解题思路

分母有三个不同的线性因子，逐个代入对应零点。

SOLUTION

Clearing $2x(x-2)(x+2)$ gives the coefficients $-3/8, 5/16, 1/16$ for $1/x, 1/(x-2), 1/(x+2)$. Thus the integral equals

$$-\frac{3}{8}\ln|x| + \frac{5}{16}\ln|x-2| + \frac{1}{16}\ln|x+2| + C.$$

参考教材例题 2，第 491 页

Exercise 20

解题思路

重根因子的一次和二次倒数都要列出。

SOLUTION

Clearing denominators and matching coefficients gives

$$\frac{1}{4(x-1)} + \frac{3}{4(x+1)} - \frac{1}{2(x+1)^2}.$$

Integration yields $\frac{1}{4}\ln|x-1| + \frac{3}{4}\ln|x+1| + 1/[2(x+1)] + C$.

参考教材例题 5，第 493 页

Exercise 23

解题思路

分子拆成一个完整的二次因子与该因子的导数。

SOLUTION

The integrand is $1/(y^2+1) + 2y/(y^2+1)^2$. The integral is $\arctan y - 1/(y^2+1) + C$.

参考教材例题 4，第 492 页

Exercise 27

解题思路

拆出线性因子，再把二次分母的分子分为导数项和常数项。

SOLUTION

Factor $x^3-1=(x-1)Q$ with $Q=x^2+x+1$. Clearing denominators gives $2/[3(x-1)]+(x-4)/(3Q)$. The second numerator is $Q'/6-3/2$. Completing the square in Q gives

$$\frac{2}{3}\ln|x-1|+\frac{1}{6}\ln Q-\sqrt{3}\arctan\frac{2x+1}{\sqrt{3}}+C.$$

参考教材例题 5，第 493 页

Exercise 31

解题思路

将分子写成二次因子的倍数加上它的导数。

SOLUTION

Put $Q=u^2+2u+2$. The integrand is $(2u+1)/Q+(2u+2)/Q^2$, or $(Q'-1)/Q+Q'/Q^2$. Since $Q=(u+1)^2+1$, the integral is $\ln Q-\arctan(u+1)-1/Q+C$.

参考教材例题 3，第 492 页

Exercise 34

解题思路

先做多项式除法，剩余真分式再拆。

SOLUTION

The quotient and remainder give $x^2+1+1/(x^2-1)$. The last term is $\frac{1}{2}[1/(x-1)-1/(x+1)]$. Therefore the integral is $x^3/3+x+\frac{1}{2}\ln|(x-1)/(x+1)|+C$.

参考教材例题 3，第 492 页；例题 2，第 491 页

Exercise 35

解题思路

除法后保留重根对应的所有项。

SOLUTION

Dividing and clearing denominators gives $9+2/x-1/x^2+7/(x-1)$. Its integral is $9x+2\ln|x|+1/x+7\ln|x-1|+C$.

参考教材例题 1，第 490 页

Exercise 41

解题思路

先令正弦为变量，把分母化成可分解的二次多项式。

SOLUTION

Set $u=\sin y$. Then $du=\cos y\,dy$ and the integrand is $du/[(u-2)(u+3)]$. Its partial fractions give $\frac{1}{5}\ln|(u-2)/(u+3)|+C$. Replace u by $\sin y$.

8.7 Numerical Integration

6 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 510–512

参考教材例题 1–3，第 504、506、508 页

Exercise 8

解题思路

先算五个节点的函数值，再分别按梯形和辛普森权重求和。误差界用二阶或四阶导数。

SOLUTION

Let $f(s) = (s-1)^{-2}$ and $h = 1/2$. The five values are $1, 4/9, 1/4, 4/25, 1/9$. The exact integral is $[-1/(s-1)]_2^4 = 2/3$.

(I a) The trapezoidal estimate is $T = 141/200 = 0.705$. Since $\max|f''| = 6$, its error is bounded by $6(2)^3/(12 \cdot 4^2) = 1/4$.

(I b–c) The actual error is $23/600$, or 5.75%.

(II a) Simpson's estimate is $S = 1813/2700 \approx 0.671481$. Since $\max|f^{(4)}| = 120$, the bound is $120(2)^5/(180 \cdot 4^4) = 1/12$.

(II b–c) The actual error is $13/2700$, or approximately 0.72222%.

参考教材例题 1，第 504 页

Exercise 24

解题思路

时间间隔不同，每段单独用梯形面积并统一单位。

SOLUTION

Include the initial pair $(0, 0)$. With speeds v_i in km/h and times t_i in seconds, the estimated distance in meters is

$$\sum_{i=1}^{11} \frac{(v_i + v_{i-1})(t_i - t_{i-1})}{7.2}.$$

The numerator sum is 7045, giving approximately 978.47 m.

参考教材例题 6，第 510 页

Exercise 25

解题思路

先用辛普森法估算横截面积，再由质量和密度求总体积。

SOLUTION

The spacing is 0.3 m. Simpson's area is $0.1[0.50 + 4(0.55 + 0.65 + 0.75) + 2(0.60 + 0.70) + 0.75] = 1.165$ m^2. The volume is $2000/673$ m^3. Hence the tank length is approximately 2.551 m.

参考教材例题 2–3，第 506、508 页

Exercise 28

解题思路

按辛普森权重积分指数函数，最后把常数乘到估计和误差界上。

SOLUTION

For (a), take $t_i = i/10$ and $f_i = e^{-t_i^2}$. Simpson's weighted sum is

$$\frac{2}{30\sqrt{\pi}}(f_0 + 4f_1 + 2f_2 + \cdots + 4f_9 + f_{10})$$

giving $\operatorname{erf}(1) \approx 0.84270171$.

For (b), the bound is $\frac{2}{\sqrt{\pi}} \frac{12}{180 \cdot 10^4} \approx 7.523 \times 10^{-6}$.

参考教材例题 1，第 504 页；介值定理

Exercise 29

解题思路

端点函数值的平均数夹在两端函数值之间，可以由区间内某一点取得。

SOLUTION

Continuity and the Intermediate Value Theorem give $c_i \in [x_{i-1}, x_i]$ with $f(c_i) = [f(x_{i-1}) + f(x_i)]/2$. Each trapezoid contributes $f(c_i)\Delta x_i$. Thus their sum is a Riemann sum for this partition.

参考教材例题 1、3，第 504、508 页

Exercise 31

解题思路

将椭圆长度写成四倍的四分之一弧长，并把误差界也乘四。

SOLUTION

For (a), use $f(t) = \sqrt{1 - \cos^2 t/4}$ and $h = \pi/20$. The length estimate is

$$4h\left[\frac{f(0) + f(\pi/2)}{2} + \sum_{i=1}^{9} f(ih)\right] \approx 5.869849.$$

For (b), the bound using $|f''| < 1$ is $4(\pi/2)^3/(12 \cdot 10^2) = \pi^3/2400 \approx 0.01292$. The printed key's value of about 0.0032 corresponds to one quadrant with this derivative bound.

8.8 Improper Integrals

19 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 521–522

参考教材例题 4，第 516 页

Exercise 4

解题思路

先把上限截在奇点左侧，积分后再取极限。

SOLUTION

For $b < 4$, the integral to b is $4 - 2\sqrt{4-b}$. As $b \to 4^-$ it tends to 4.

参考教材例题 2，第 514 页

Exercise 10

解题思路

截断负无穷，用反正切原函数。

SOLUTION

The antiderivative is $\arctan(x/2)$. As the lower endpoint tends to $-\infty$, the integral tends to $\pi/4 - (-\pi/2) = 3\pi/4$.

参考教材例题 2，第 514 页

Exercise 14

解题思路

全实轴积分要先证明左右两边各自收敛。

SOLUTION

An antiderivative is $-1/\sqrt{x^2+4}$. The integral from $-\infty$ to 0 is $-1/2$ and the integral from 0 to ∞ is $1/2$. The sum is 0.

参考教材例题 2，第 514 页

Exercise 20

解题思路

反正切既是换元变量，也在分母对应的微分里。

SOLUTION

The antiderivative is $8(\arctan x)^2$. Its limiting increment from 0 to ∞ is $8(\pi/2)^2 = 2\pi^2$.

参考教材例题 2，第 514 页

Exercise 23

解题思路

积分区间只在非正半轴，先去掉绝对值。

SOLUTION

For $x \le 0$, $e^{-|x|} = e^x$. The integral from $a < 0$ to 0 is $1 - e^a$, tending to 1 as $a \to -\infty$.

参考教材例题 4，第 516 页

Exercise 27

解题思路

在根式为零前截断，再取反正弦的端点极限。

SOLUTION

The integral from 0 to $b < 2$ is $\arcsin(b/2)$. Letting $b \to 2^-$ gives $\pi/2$.

参考教材例题 5，第 517 页

Exercise 32

解题思路

在内部奇点 1 分开，两侧分别取极限。

SOLUTION

The two one-sided integrals are $\int_0^1 (1-x)^{-1/2}\,dx = 2$ and $\int_1^2 (x-1)^{-1/2}\,dx = 2$, each interpreted as an endpoint limit. The sum is 4.

参考教材例题 4，第 516 页

Exercise 39

解题思路

正切积分是负的余弦对数，余弦趋于零时发散。

SOLUTION

The truncated integral is $-\ln(\cos b)$ for $b < \pi/2$. It tends to $+\infty$ as $b \to \pi/2^-$, so the integral diverges.

参考教材例题 4，第 516 页

Exercise 43

解题思路

把指数内的负倒数看作新变量，它的导数正好是平方倒数。

SOLUTION

The integrand has antiderivative $e^{-1/x}$. This tends to zero as $x \to 0^+$. Thus the integral converges to $e^{-1/\ln 2}$.

参考教材例题 7，第 519 页

Exercise 45

解题思路

在零附近，分母的根号项占主导，用幂函数作比较。

SOLUTION

On $(0, \pi]$, $\sin t \geq 0$, so $0 < 1/(\sqrt{t} + \sin t) \leq t^{-1/2}$. The comparison integral is finite. Hence the given integral converges.

参考教材例题 7，第 519 页

Exercise 56

解题思路

在无穷远处和调和型积分作比较。

SOLUTION

The ratio of $1/\sqrt{x^2-1}$ to $1/x$ is $x/\sqrt{x^2-1} \to 1$. Since $\int_2^\infty dx/x$ diverges, the given integral diverges.

参考教材例题 7，第 519 页

Exercise 57

解题思路

根号分子至多按平方根的速度增长，整体用负二分之三次幂比较。

SOLUTION

For $x \geq 1$, $\sqrt{x+1}/x^2 \leq \sqrt{2}\,x^{-3/2}$. Since $3/2 > 1$, the comparison integral converges, hence so does the given integral.

参考教材例题 6，第 518 页

Exercise 60

解题思路

利用正弦有界，把被积函数夹在零和平方倒数之间。

SOLUTION

We have $0 \leq (1+\sin x)/x^2 \leq 2/x^2$ for $x \geq \pi$. The upper comparison integral converges, so the given integral converges.

参考教材例题 6，第 518 页

Exercise 65

解题思路

用指数的增长控制根号分母。

SOLUTION

For $x \geq 1$, $x/e^x \leq 1/e$, since xe^{-x} is nonincreasing there. Thus $e^x - x \geq (1 - 1/e)e^x$ and the integrand is bounded by $(1-1/e)^{-1/2}e^{-x/2}$. The latter is integrable, so the integral converges.

参考教材例题 6，第 518 页

Exercise 68

解题思路

被积函数是偶函数，只需比较右半轴的指数尾部。

SOLUTION

The integrand is even. For $x \geq 0$, $1/(e^x + e^{-x}) \leq e^{-x}$. Hence both tails converge by comparison.

参考教材例题 3，第 515 页；例题 4，第 516 页

Exercise 69

解题思路

令自然对数为新变量，两个问题分别变为零点和无穷远处的幂积分。

SOLUTION

With $u = \ln x$, (a) becomes $\int_0^{\ln 2} u^{-p}\,du$, converging exactly for $p < 1$. Part (b) becomes $\int_{\ln 2}^{\infty} u^{-p}\,du$, converging exactly for $p > 1$. The case $p = 1$ diverges at both ends.

参考教材例题 1，第 514 页

Exercise 71

解题思路

面积是指数函数从零到无穷的正积分。

SOLUTION

The area through $x = b$ is $\int_0^b e^{-x}\,dx = 1 - e^{-b}$. Its limit is 1.

参考教材例题 4，第 516 页；6.4 节例题 1，第 358 页

Exercise 76

解题思路

垫圈用平方差化简。表面积另用半径乘弧长，比较一个已经发散的积分。

SOLUTION

For (a), $\sec x \geq \tan x \geq 0$ on $[0, \pi/2)$ and $\sec^2 x - \tan^2 x = 1$. Thus $V = \pi \int_0^{\pi/2} dx = \pi^2/2$.

For (b), each lateral area is $\int 2\pi y\sqrt{1 + (y')^2}\,dx$, bounded below by $\int 2\pi y\,dx$. For $y = \tan x$ this lower bound diverges by Exercise 39. It also diverges for $y = \sec x \geq \tan x$, so both areas are infinite.

参考教材例题 2，第 514 页；8.4 节例题 4，第 487 页

Exercise 80

解题思路

用正割换元同时处理下限奇点与无穷远。

SOLUTION

Set $x = 3\sec\theta$ on $0 < \theta < \pi/2$. The truncated integrand becomes $d\theta/3$. As the endpoints tend to 3^+ and ∞, the angles tend to 0 and $\pi/2$. Hence the integral is $\pi/6$.

16.1 Solutions, Slope Fields, and Euler's Method

7 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 1022–1023

参考教材例题 3，第 1019 页

Exercise 1

解题思路

在图中找零斜率的位置，再用对角线两侧的正负区分。

SOLUTION

This is field (d). The slopes vanish on $y = -x$, are positive above that line, and are negative below it.

参考教材例题 3，第 1019 页

Exercise 2

解题思路

斜率只依赖纵坐标，同一横行方向一致。

SOLUTION

This is field (c). Slopes are constant along horizontal rows and vanish at $y = -1$. They are positive above and negative below that row.

参考教材例题 3，第 1019 页

Exercise 3

解题思路

看切线是否与指向原点的半径垂直。

SOLUTION

This is field (a). Where $y \neq 0$, the slope $-x/y$ is tangent to the circle $x^2+y^2 = C$. The direction field is undefined on $y = 0$.

参考教材例题 3，第 1019 页

Exercise 4

解题思路

斜率在两条对角线上都为零，对角线之间分区判断。

SOLUTION

This is field (b). The slopes vanish on $y = \pm x$, are positive where $|y| > |x|$, and are negative where $|y| < |x|$.

参考教材例题 3，第 1019 页；图 16.3–16.4

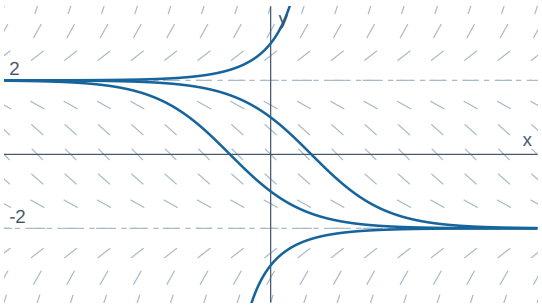
Exercise 5

解题思路

先画两条平衡线，再按各带的斜率方向穿过短线。

SOLUTION

The equilibrium curves are $y = \pm 2$. Curves below -2 increase toward -2, curves between -2 and 2 decrease toward -2, and curves above 2 increase away from 2. The plotted curves follow the field in each strip.



参考教材例题 3，第 1019 页

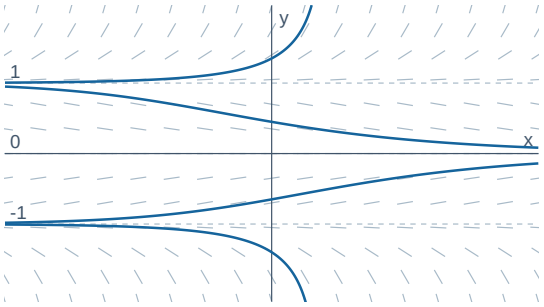
Exercise 6

解题思路

三个平衡值把方向场分成四个横带，分别判断斜率正负。

SOLUTION

The equilibria are $y = -1, 0, 1$. From bottom to top, the signs of y' are $-, +, -, +$. Thus solutions on either side of zero within $(-1, 1)$ approach zero forward in time, while the outer solutions move away from the adjacent equilibrium.



参考教材例题 4–5，第 1020–1021 页

Exercise 18

解题思路

先按欧拉递推做三步，再分离变量求精确解并逐点比较。

SOLUTION

With $x_0 = -1$, $y_0 = 1$ and $h = 0.5$, use $y_{k+1} = y_k + 0.5y_k^2(1 + 2x_k)$. At $x = -0.5, 0, 0.5$ the estimates are $0.5000, 0.5000, 0.6250$.

Separating gives $-1/y = x + x^2 + C$. The initial condition gives $C = -1$, hence $y = 1/(1 - x - x^2)$. At the three nodes the exact values are $0.8000, 1.0000, 4.0000$. The absolute errors are $0.3000, 0.5000, 3.3750$, all underestimates.

7.4 Exponential Change and Separable Differential Equations

7 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 423–425

参考教材例题 1，第 417 页

Exercise 11

解题思路

把指数差拆成乘积，把含因变量的因子移到左边。

SOLUTION

From $y' = e^x e^{-y}$ we get $e^y\,dy = e^x\,dx$. Thus $e^y = e^x + C$, or $y = \ln(e^x + C)$ where the argument is positive.

参考教材例题 1，第 417 页

Exercise 12

解题思路

乘以因变量的指数后，左侧成为一个全导数。

SOLUTION

Since $(e^y)' = e^y y' = 3x^2$, integration gives $e^y = x^3 + C$. Hence $y = \ln(x^3 + C)$ where defined.

参考教材例题 1，第 417 页

Exercise 13

解题思路

令新变量为因变量的平方根，再分离变量。另检查分离时除去的平衡解。

SOLUTION

For $y > 0$, put $u = \sqrt{y}$. Then $2u' = \cos^2 u$. Where $\cos u \neq 0$, integration yields $\tan u = x/2 + C$, with $u \geq 0$ on a continuous branch. Thus $y = u^2$. The constant solutions are 0 and $(\pi/2 + k\pi)^2$ for $k \geq 0$. A branch reaching $u = 0$ can be joined to $y = 0$ on its left, since $y' = 2uu' \to 0$ there.

参考教材例题 2，第 418 页

Exercise 17

解题思路

分离后左边是反正弦的微分，注意它的主值区间。

SOLUTION

On $-1 < y < 1$, separation gives $\arcsin y = x^2 + C$. Thus $y = \sin(x^2 + C)$ on intervals where $-\pi/2 < x^2 + C < \pi/2$.

参考教材例题 3，第 418 页

Exercise 30

解题思路

两个时刻的人口比值给出两小时的增长倍率。

SOLUTION

For $P = P_0 e^{kt}$, $e^{2k} = 40000/10000 = 4$, so $e^k = 2$. Hence $P_0 = 10000/2^3 = 1250$ bacteria.

参考教材例题 3，第 418 页

Exercise 34

解题思路

用初始瞬时增长率除以初始人口得到相对增长率。

SOLUTION

The initial absolute rate is $365(86400)/12 = 2628000$ people/year. Therefore (a) $k = 2628000/314419198 \approx 0.00835827$ per year.

(b) At seven model years, $P = 314419198e^{7k} \approx 333364004$ people.

参考教材例题 6，第 422 页

Exercise 42

解题思路

用两次测温相对于室温的差值，求每十分钟的衰减倍率。

SOLUTION

Write $T(t)-18=Ce^{-kt}$. The deviations at 10 and 20 min are -16 and -8, so $e^{-10k}=1/2$. Hence $C=-32$ and $T(0)=-14$°C.

16.2 FirstOrder Linear Equations

4 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 1028–1029

参考教材例题 2，第 1026 页

Exercise 1

解题思路

左边已经是横坐标与因变量的乘积导数。

SOLUTION

Because $(xy)' = xy' + y = e^x$, integration gives $xy = e^x + C$. Thus $y = (e^x + C)/x$ for $x > 0$.

参考教材例题 2，第 1026 页

Exercise 4

解题思路

先求积分因子，把方程左边合成一个乘积导数。

SOLUTION

On the stated interval, $\cos x > 0$ and the integrating factor is $\sec x$. Multiplication gives $(y \sec x)' = \cos x$, hence $y = \cos x(\sin x + C)$.

参考教材例题 3，第 1027 页

Exercise 16

解题思路

化成标准形式后积分因子是时间的平方，最后用初值。

SOLUTION

After division by t, the integrating factor is t^2. Thus $(t^2 y)' = t^4$ and $t^2 y = t^5/5 + C$. At $t = 2$, $C = 4 - 32/5 = -12/5$. Hence $y = t^3/5 - 12/(5t^2)$.

参考教材例题 3，第 1027 页

Exercise 17

解题思路

直接识别乘积导数，再用给定角度处的值求常数。

SOLUTION

Since $(uy)' = \sin u$, we have $uy = -\cos u + C$. The condition $y(\pi/2) = 1$ gives $C = \pi/2$, so $y = (\pi/2 - \cos u)/u$.

16.3 Applications

2 assigned exercises • Thomas' Calculus, 14th SI edition, p. 1036

参考教材例题 3，第 1033 页

Exercise 13

解题思路

建立盐量收支方程。体积随时间增加，流出浓度用瞬时盐量除以瞬时体积。

SOLUTION

Let $A(t)$ be the salt mass in kg.

(a) The inflow is $0.2(20) = 4$ kg/min.

(b) The volume is $400 + 4t$ L.

(c) The salt outflow is $16A/(400 + 4t)$ kg/min.

(d) Thus $A' + 4A/(100 + t) = 4$, with $A(0) = 8000$. The integrating factor is $(100 + t)^4$. Integrating and using the initial mass gives

$$A = 0.8(100 + t) + 7920\left(\frac{100}{100 + t}\right)^4.$$

(e) At 25 min the concentration is $0.2 + 19.8(4/5)^5 = 6.688064$ kg/L.

参考教材例题 3，第 1033 页

Exercise 15

解题思路

先解肥料总量的方程，再在排空前找极大值。

SOLUTION

Let A be the fertilizer mass in kg and $u = 50 - t > 0$. The volume is $8u$ L, so $A' = 0.4 - 3A/(2u)$ and $A(0) = 0$. In the variable u, multiplying $dA/du - 3A/(2u) = -0.4$ by $u^{-3/2}$ gives

$$A = 0.8u(1 - \sqrt{u/50}).$$

Its derivative with respect to u is $0.8[1 - \frac{3}{2}\sqrt{u/50}]$, zero at $u = 200/9$. The endpoints give zero mass and this is the unique maximum. Thus the maximum is $160/27$ kg at $t = 250/9$ min.

16.4 Graphical Solutions of Autonomous Equations

6 assigned exercises • Thomas' Calculus, 14th SI edition, pp. 1042–1043

参考教材例题 1，第 1037 页

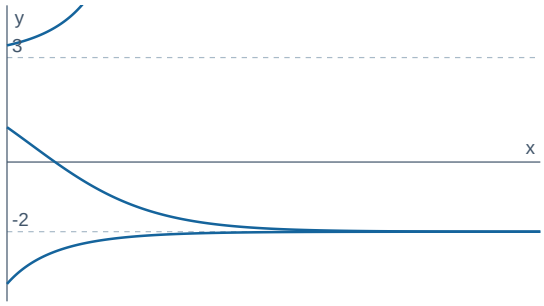
Exercise 1

解题思路

标出两个平衡点，再用一阶导数的符号画相线和解曲线。

SOLUTION

The equilibria are -2 and 3. In increasing order of y, the phase-line arrows are $\rightarrow$ below -2, $\leftarrow$ on $(-2, 3)$, and $\rightarrow$ above 3. Thus -2 is stable and 3 is unstable. The curves below 3 approach -2 forward in time, while curves above 3 rise away from it.



参考教材例题 1，第 1037 页

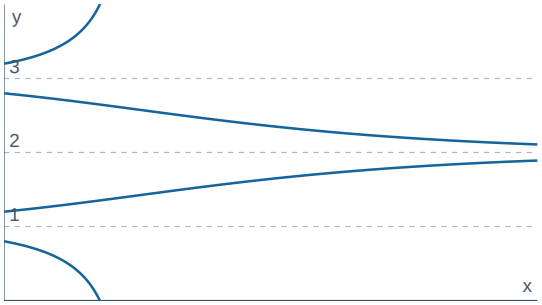
Exercise 7

解题思路

把三个线性因子的符号相乘，判断相邻解是靠近还是离开平衡线。

SOLUTION

The equilibria are $1, 2, 3$. The signs of y' from bottom to top are $-, +, -, +$, so 1 and 3 are unstable and 2 is stable. Curves between 1 and 3 approach 2 forward in time, while curves outside move away from the nearest equilibrium.



参考教材图 16.26，第 1042 页；例题 1，第 1037 页

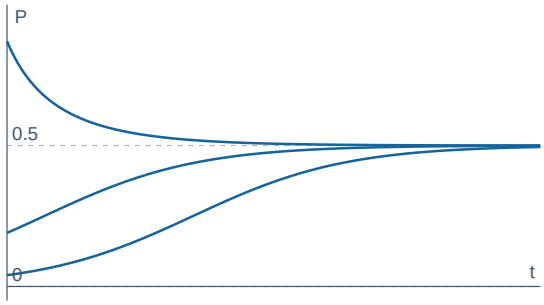
Exercise 10

解题思路

在非负人口区间，依据增长率的符号判断承载量。

SOLUTION

The equilibria are 0 and $1/2$. For $0 < P < 1/2$, $P' > 0$, and for $P > 1/2$, $P' < 0$. Hence 0 is unstable and $1/2$ is stable. Positive solutions approach $1/2$ from below or above according to their initial value.



参考教材例题 1，第 1037 页；图 16.26，第 1042 页

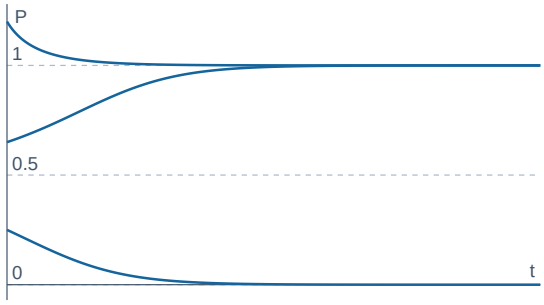
Exercise 12

解题思路

中间平衡值是阈值，阈值两侧的箭头指向不同稳态。

SOLUTION

The equilibria are $0, 1/2, 1$. For P in $(0, 1/2)$, $(1/2, 1)$ and $(1, \infty)$ the signs are $-, +, -$, respectively. Thus 0 and 1 are stable and $1/2$ is unstable. Subthreshold positive solutions decrease to 0, while those above the threshold approach 1.



参考教材例题 1，第 1037 页；图 16.26，第 1042 页

Exercise 14

解题思路

灭绝阈值和承载量一起决定增长方向。用单调有界与平衡条件确定极限。

SOLUTION

Assume $0 < m < M$.

(a) The phase line on $P \geq 0$ has equilibria $0, m, M$ and signs $-, +, -$ on the three positive intervals. It models decline below m, recovery above m, and decline above M.

(b) The logistic model has positive growth at all $0 < P < M$. The added threshold represents the stated extinction behavior and makes this model more suitable for that premise.

(c) Above M, the solution decreases and is bounded below by M. If its limit were greater than M, continuity would keep its derivative bounded away from zero on the negative side, a contradiction. Hence the limit is M.

(d) A positive solution below m decreases to 0 by the same argument.

(e) Initial values $0, m, M$ give constant solutions. Below m the steady state is 0, above m it is M. For a one-time issue at population $P_0 > m$, keep $P_0 - H > m$. If $P_0 > M$, removing roughly $P_0 - M$ deer brings the stock near capacity. The model gives no single permit count without a current population and management target.

参考教材图 16.26，第 1042 页；例题 1，第 1037 页

Exercise 18

解题思路

把信息传播看成逻辑斯蒂增长。用相线画曲线，再把增长率当作关于人数的二次式求最大值。

SOLUTION

For (a), potential informed–uninformed contacts scale with $X(N-X)$, and the rate vanishes if either group is empty.

(b) On $0<X<N$, $X^{'}>0$. The equilibria are $0, N$. Also $X^{''}=k(N-2X)X^{'}$, positive below $N/2$ and negative above it.

(c) For $0<X_0<N$, separation yields $X(t)=N/[1+(N/X_0-1)e^{-kNt}]$. These increasing curves approach N. The endpoint initial values give horizontal solutions.

(d) The quadratic rate is maximal at $X=N/2$. If anyone is initially informed, eventually $X\to N$. If nobody is informed, X remains 0.

